



ScopeAngleChecker

User Manual

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01

Introduction

What This App Does

- Just photograph the tip of the endoscope — the bending angle is measured automatically
- Records are saved by hospital, inspector, and scope model
- Results are compared with the specified angle automatically and marked Pass or Needs Attention
- View trends on a graph and export reports as PDF or CSV

IMPORTANT

Please Read First (Important Notes)

- You must keep using the angle measurement plate. Checking with this app does not replace checking with the plate.
- Measured values may deviate depending on shooting conditions (lighting, camera distance and angle).
- This app is not a medical device or diagnostic-support software under Japan's Pharmaceuticals and Medical Devices Act.
- Interpreting the results, and deciding whether to keep using or withdraw an endoscope, is the user's own responsibility.
- Deleting the app, changing phones, or updating iOS can erase your data. Please back up regularly to CSV or PDF.

02

First-Time Setup

What Happens on First Launch

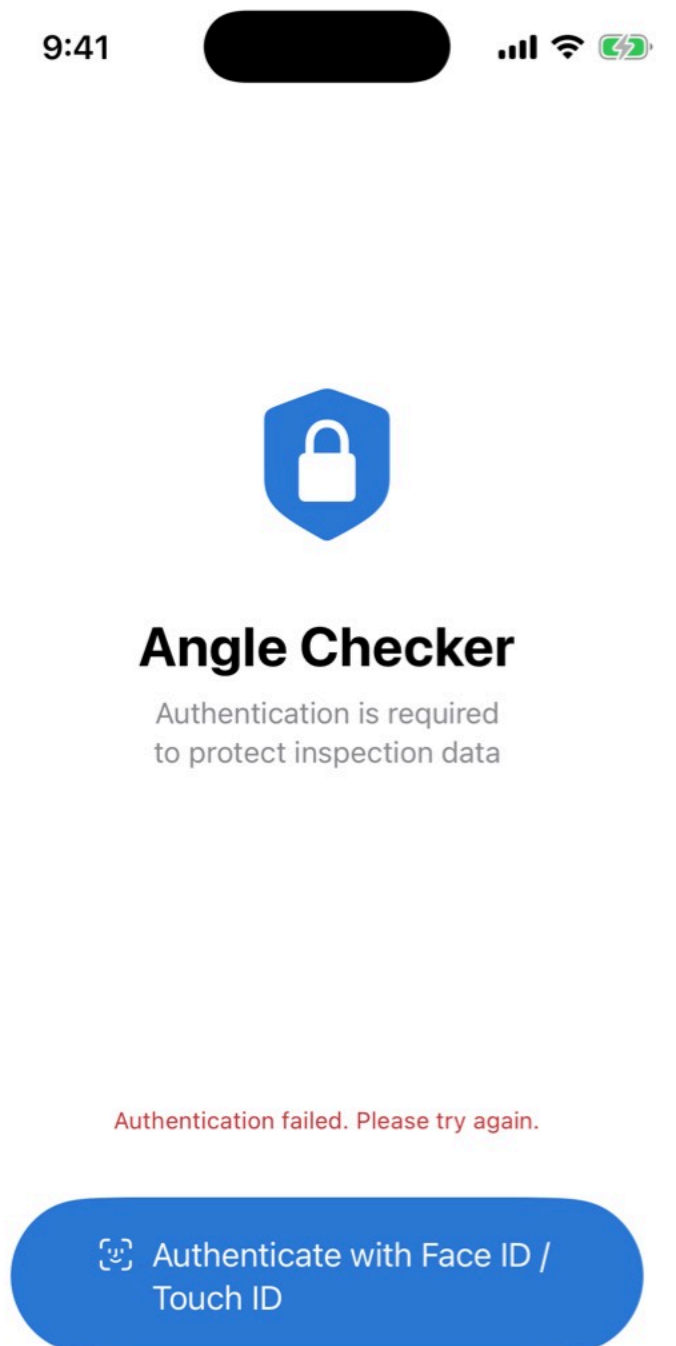
- The first time you open the app, the screens appear in this order.
- The order is fixed; you cannot skip a step.
- 1. Agree to the notes
- 2. How-to tutorial
- 3. Identity check (Face ID / Touch ID)
- 4. Data transfer prompt (choose “Using it for the first time” if this is new)
- 5. Choose a hospital / facility, then an inspector
- → Main screen (five tabs)

NOTE

About the Identity Check

- Face ID / Touch ID starts automatically when the screen opens.
- If it fails, tap “Authenticate with Face ID / Touch ID” to try again.
- If the device has no passcode at all, “Data Protection Is Disabled” appears, but you can still continue by choosing “Continue Anyway”.
- [Note] You are asked to authenticate every time you switch to another app and come back.

This is intentional, to protect your data — not a malfunction.



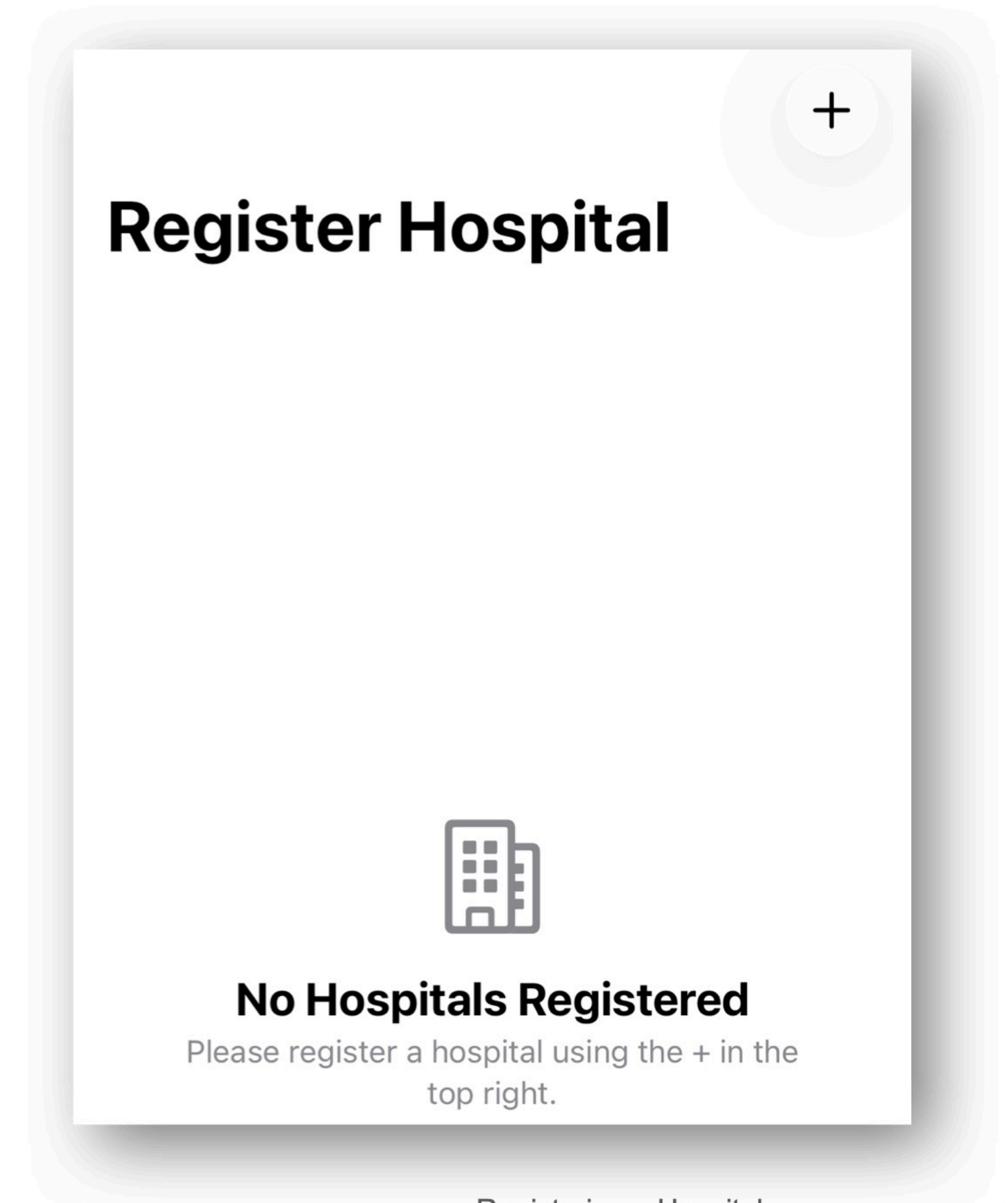
03

Hospitals and Inspectors

STEP 1

Register a Hospital / Facility

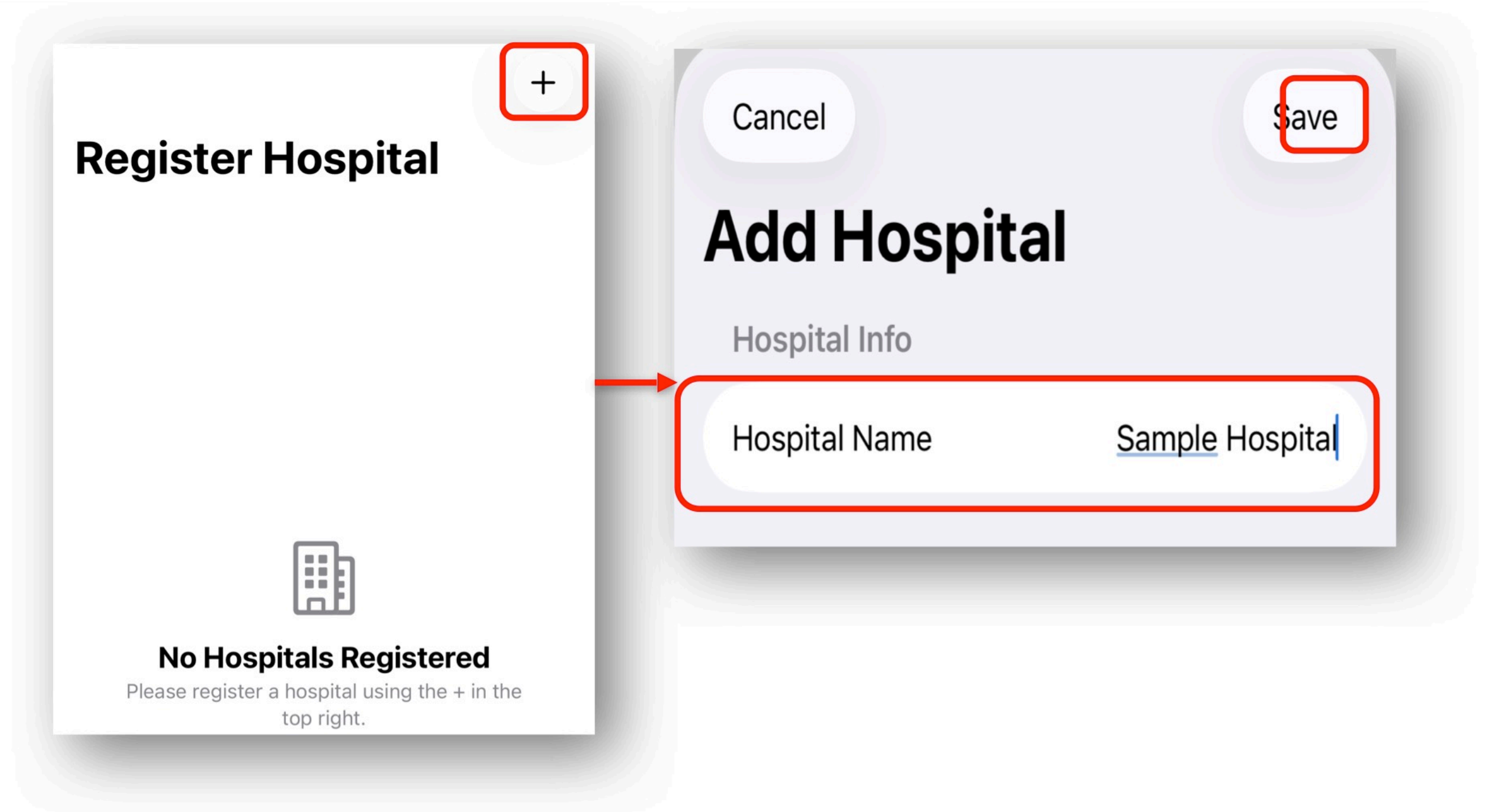
- The first time, no hospital is registered yet, so this screen appears.
- Tap “+” at the top right to add one.
- If a hospital is already registered, just tap it in the list.



STEP 2

Add a Hospital

- The only field is “Hospital Name”.
- Then tap “Save” at the top right.

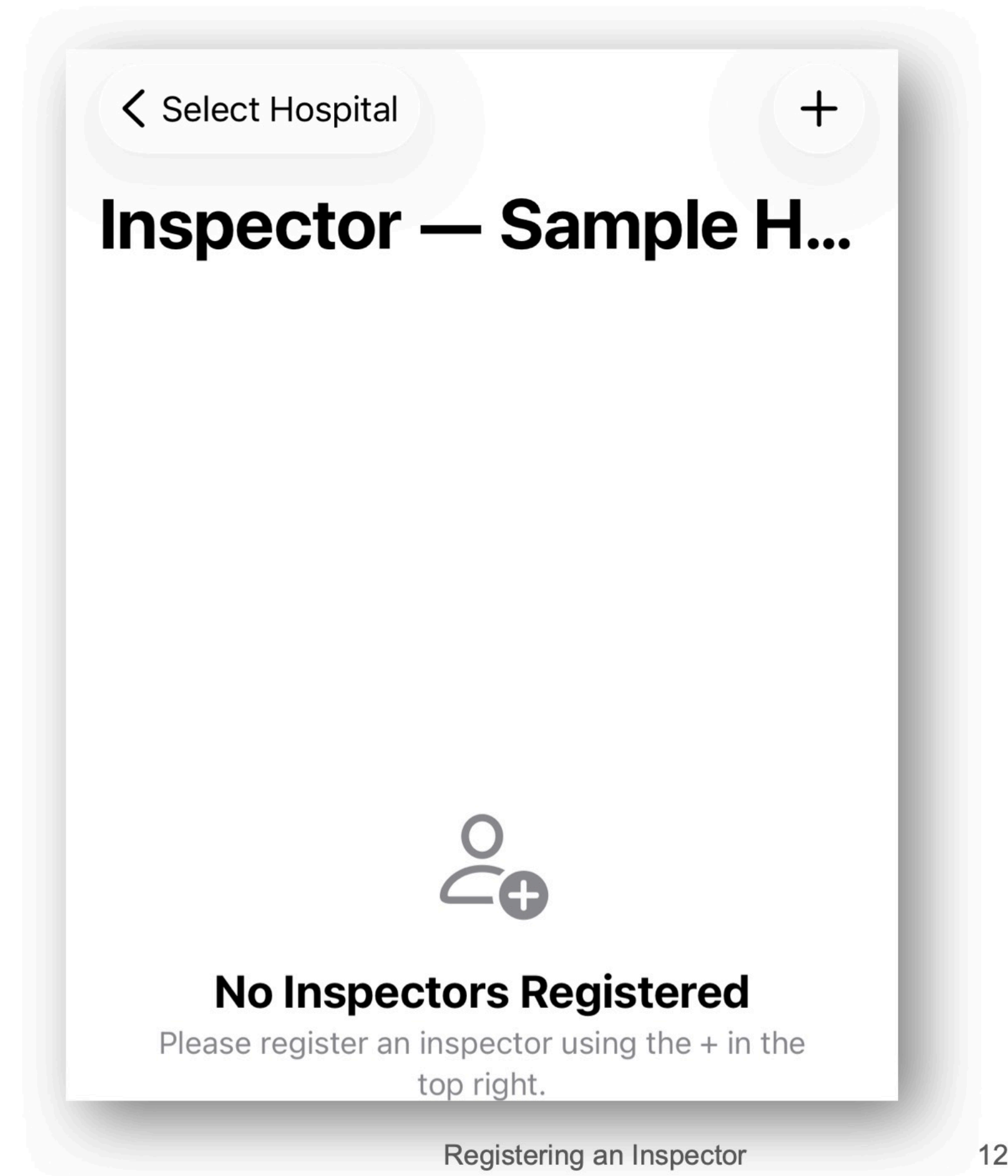


* You can register a clinic or any other facility name — it does not have to be a hospital.

STEP 3

Choose the Inspector (You)

- After you choose a hospital, the inspector screen appears next.
 - The first time, no inspector is registered yet, so this screen appears.
 - Tap “+” at the top right and enter a name.
-
- [Note] The inspector list is shared across the whole app, not kept per hospital. The same list appears whichever hospital you choose.



04

Registering Models

Registering a Model Takes Two Stages

- Registering an endoscope takes two stages.

- 1. Register the model

The specification of that model number (specified angle, tolerance, and so on). Shared by every hospital and facility.

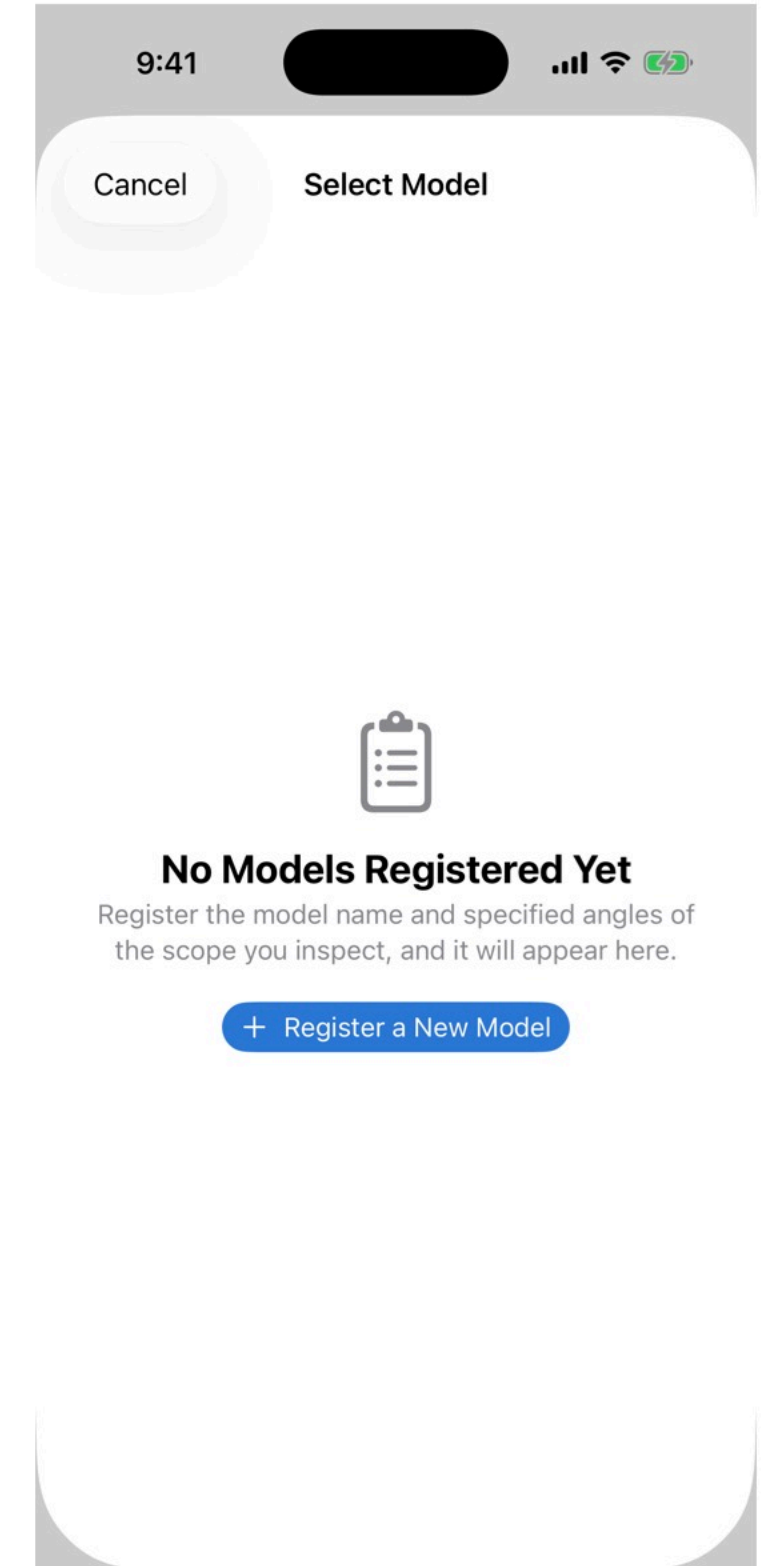
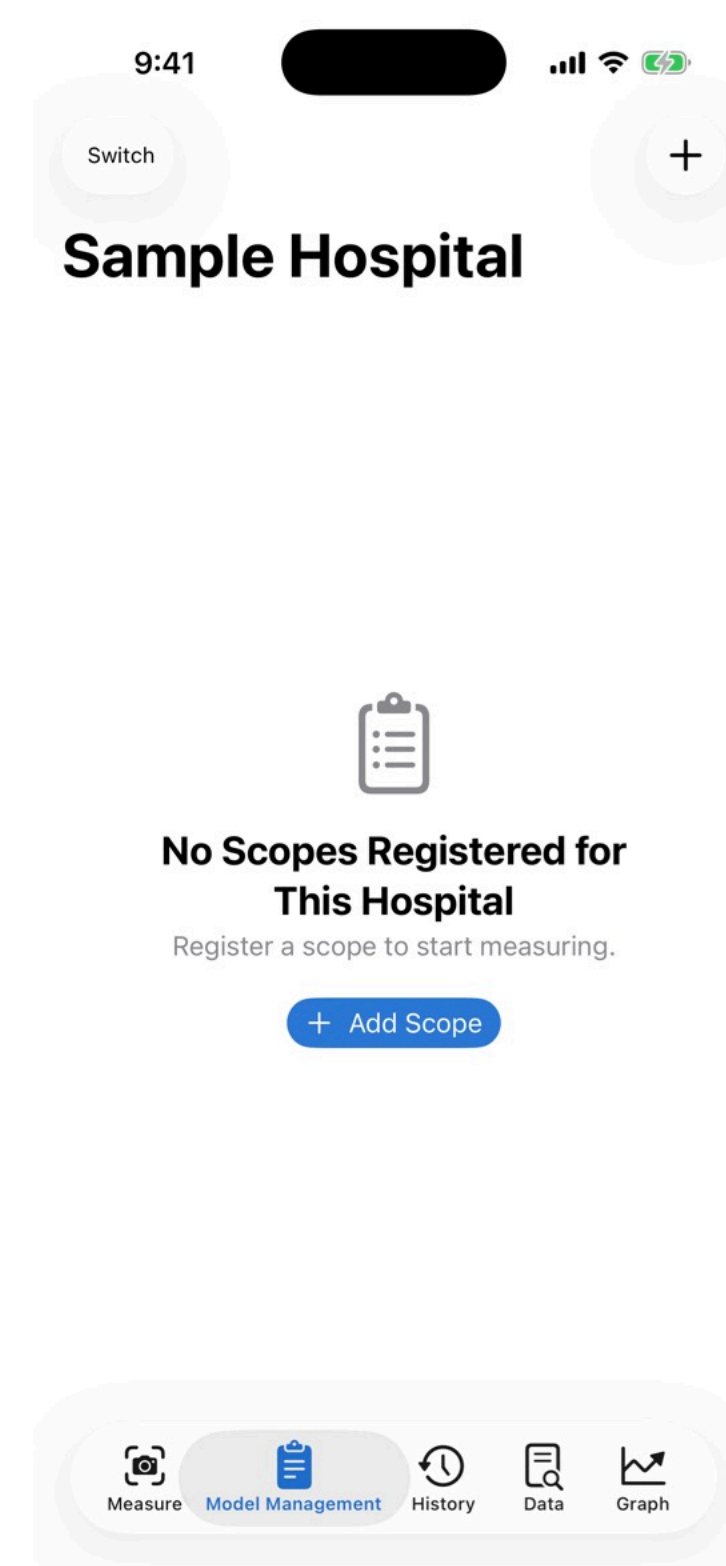
- 2. Add it as a scope at your hospital / facility

The details of the actual unit at that hospital (asset number, serial number, and so on).

STEP 1-1

Choosing or Adding a Model

- Open the “Model Management” tab and tap “+” at the top right.
- Under “Select Model”, choose from the models already registered.
- If there are none yet, tap “+ Register a New Model” to create one.
- You can do the same from Settings (gear) → “Models”.
- This is only needed the first time for a model number. After that you just select it.



STEP 1-2

Entering the Model's Specification

- Tapping “+ Register a New Model” opens this screen.
- Model Name (e.g. XYZ-1000) * required
- Manufacturer (optional) / Category (optional)
- Specified Angle: UP / DOWN / LEFT / RIGHT (optional)
- Tolerance: how many degrees from the specified angle still counts as a pass (default $\pm 30^\circ$)
- Tap “Save” at the top right; you then move straight on to entering the actual unit.

The screenshot shows the 'Register Model' screen in the ScopeAngleChecker app. At the top, there is a status bar with the time 9:41, signal strength, Wi-Fi, and battery icons. Below the status bar, there is a navigation bar with a back arrow on the left, the title 'Register Model' in the center, and a 'Save' button on the right. The main content area is divided into several sections: 'Model Info' with fields for 'Model Name' (e.g. XYZ-1000), 'Manufacturer' (Optional), and 'Category' (Not Set); 'Specified Angle (degrees)' with four options: 'UP' (Not Set), 'DOWN' (Not Set), 'LEFT' (Not Set), and 'RIGHT' (Not Set); and 'Tolerance' with a field showing '± 30 °'. Below the tolerance field, there is a small text note: 'Judged as passing if within this value of the specified angle.'

An Example of a Completed Form

- After filling it in, tap “Save” at the top right to finish.
- This example fills in the specified angle and tolerance as well.
- Once the specified angle is registered, it is used for the pass/fail judgement when measuring and for the pass band on the graph.

<

Register Model

Save

Model Info

Model Name

XYZ-1000

Manufacturer

Sample Optics

Category

Upper GI Endoscope

Specified Angle (degrees)

UP

180

DOWN

180

LEFT

160

RIGHT

160

Tolerance

±

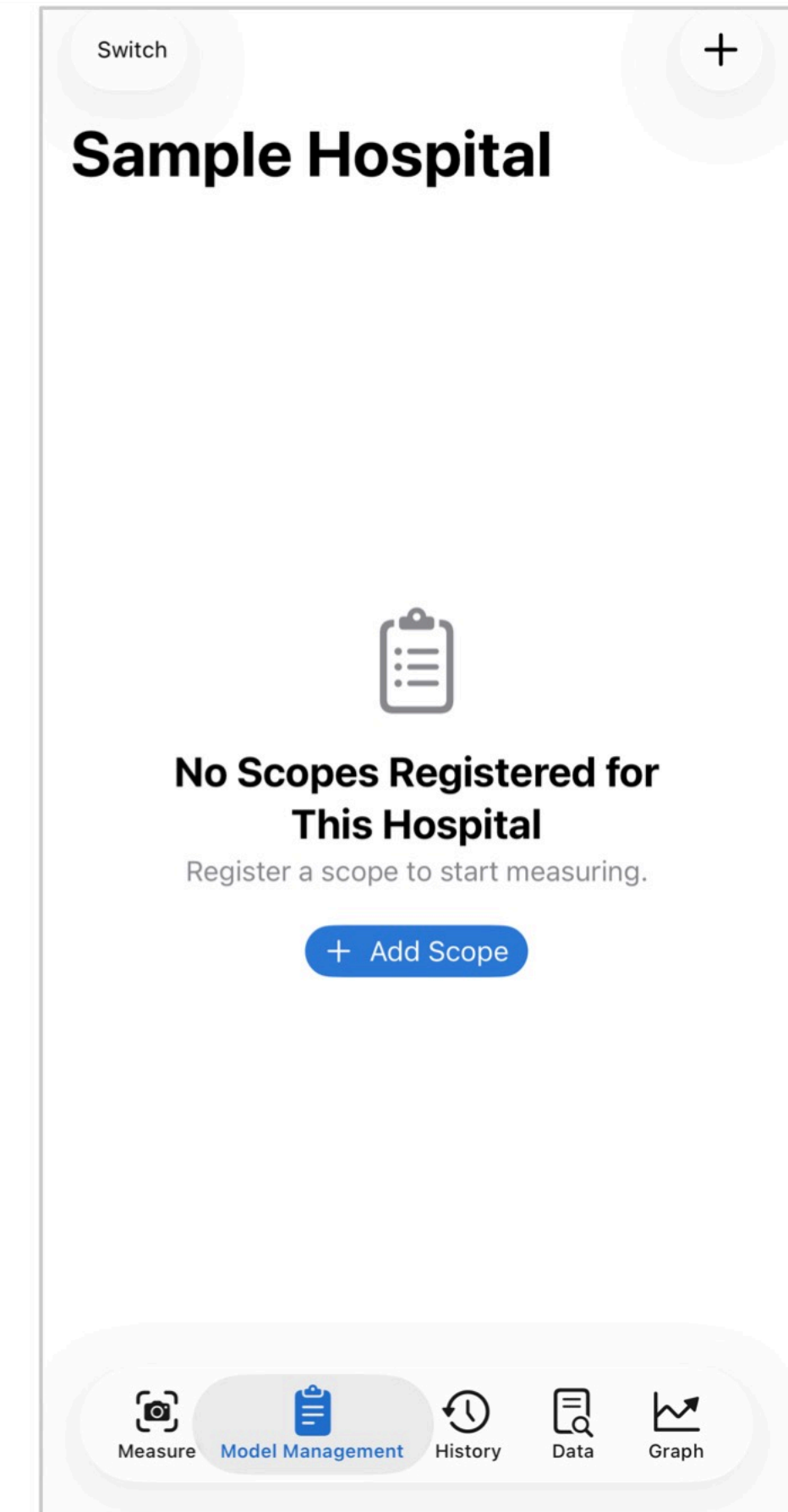
30 °

Judged as passing if within this value of the specified angle.

STEP 2-1

Add It as a Scope at the Hospital

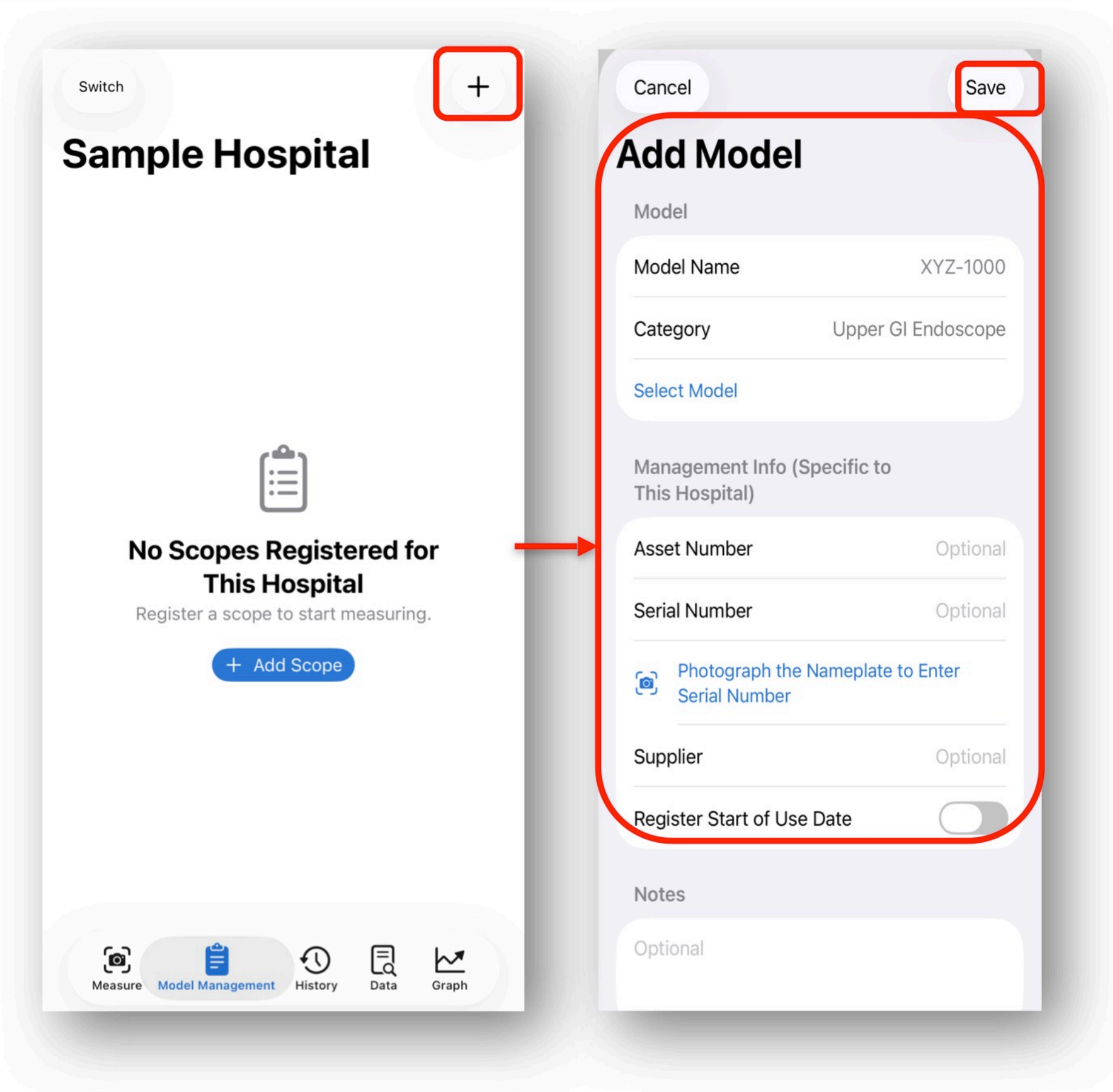
- Open the “Model Management” tab.
- No actual unit is registered at this hospital yet, so this screen appears.
- Tap “+” at the top right to register each unit.



STEP 2-2

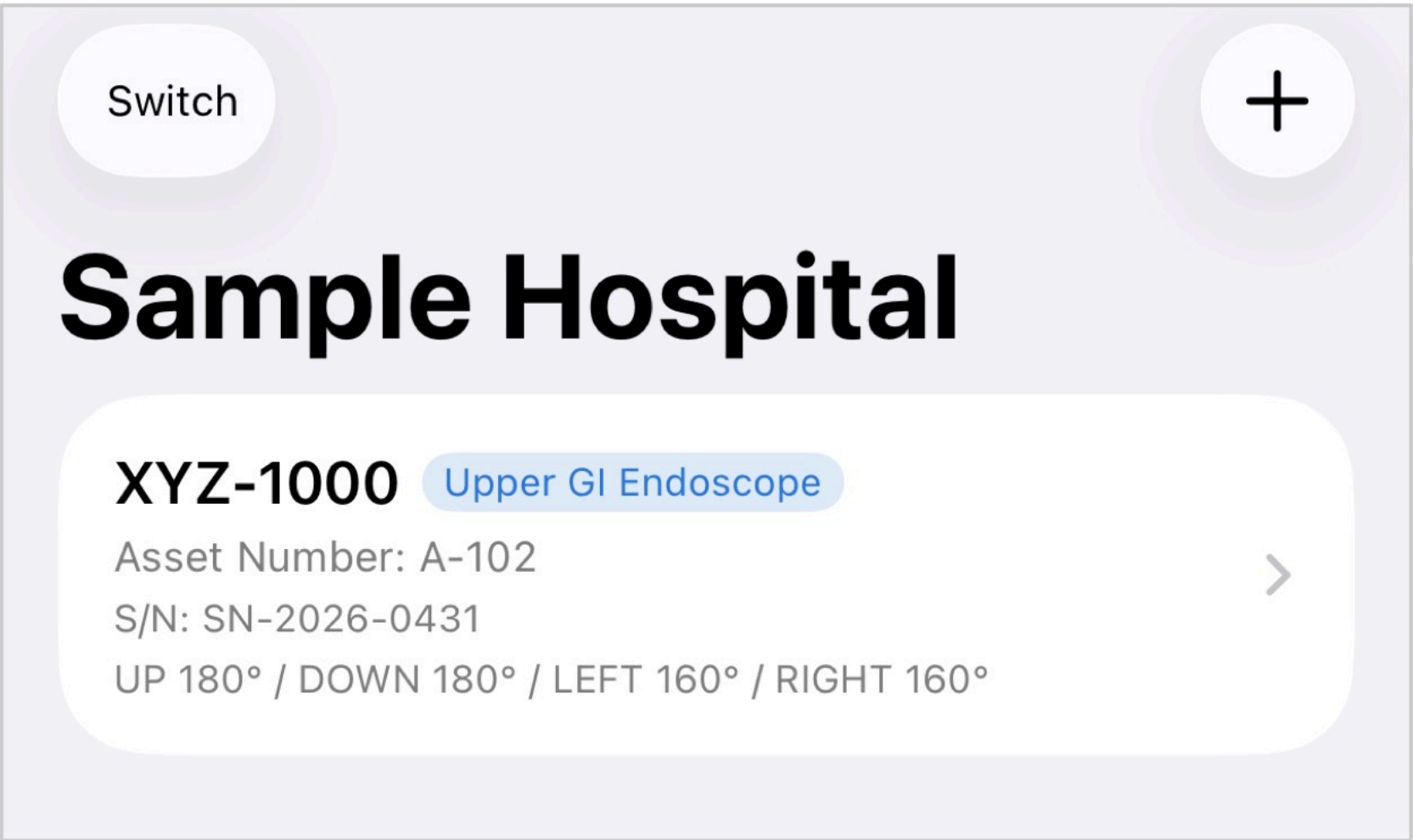
Enter the Unit's Details

- Under “Select Model”, choose the model you registered in stage 1.
- Asset Number (optional)
- Serial Number (optional): type it in, or photograph the nameplate and let the app read it (OCR, processed on the device)
- Supplier / Start of Use / Notes (all optional)



Checking What You Registered

- Once registered, the unit appears in the list on the Model Management tab.
- Model number, category, asset number, serial number, and specified angle are all visible at a glance.
- Tap an entry to edit it; swipe left to delete it.



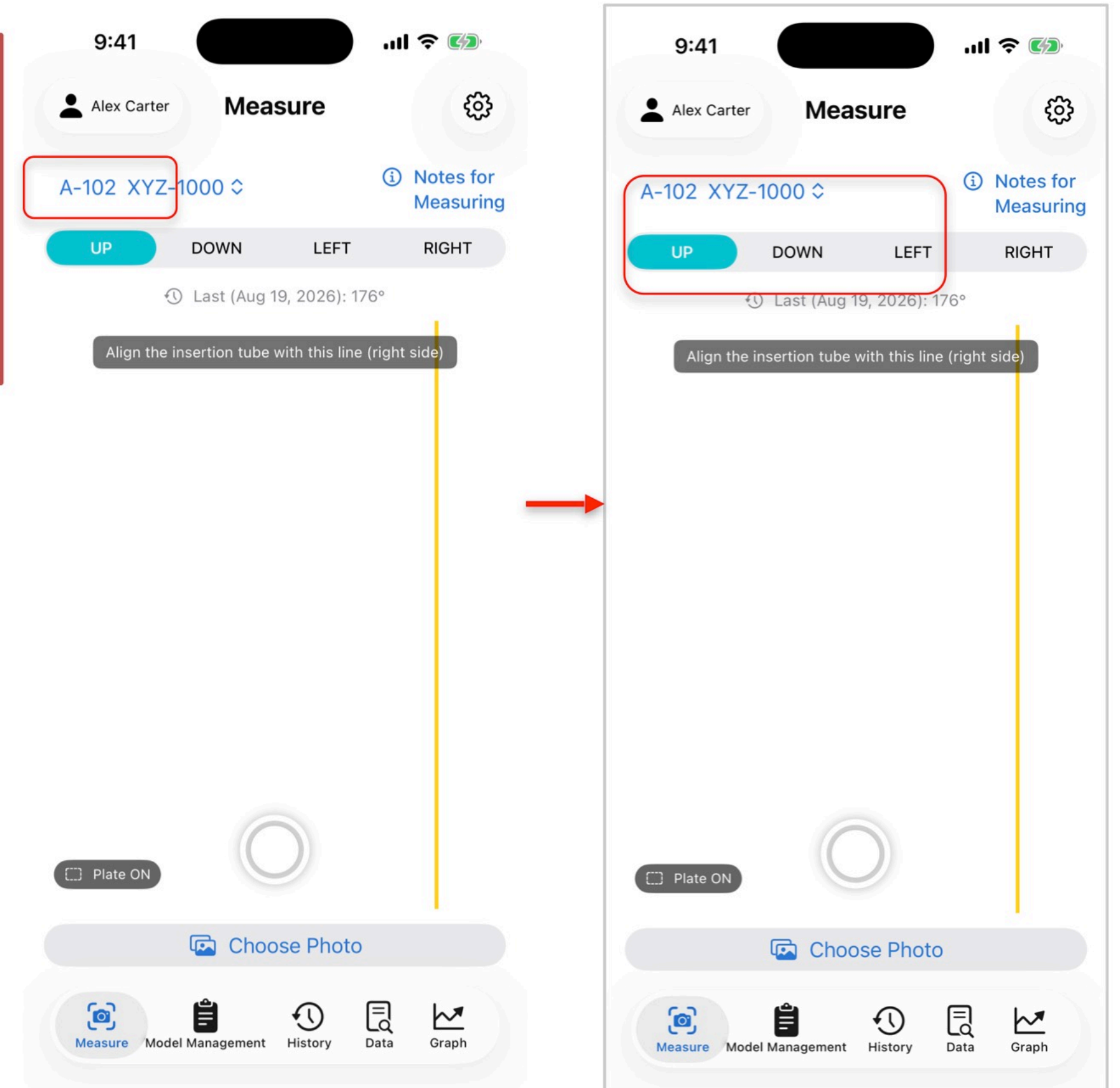
05

Measuring

STEP 1

Choose the Model and Direction

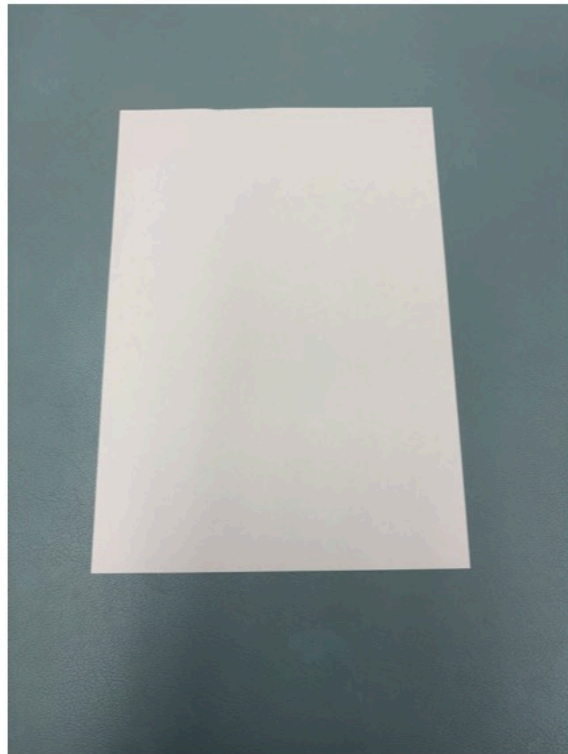
- On the “Measure” tab, choose the model and the direction (UP / DOWN / LEFT / RIGHT).
- If there are previous records for the same model and direction, the last three are shown for reference before you shoot (repeat shots on the same day count as one).



SET UP

Setting Up Your Shooting Area

- What you need: a sheet of white A4 paper / a rack with holes in its top / a simple light / sticky tape
- The paper goes under the scope; the rack holds the phone, in place of a stand.
- Aim the light so it falls evenly from above and casts no shadow. No lamp? Use “Light On” in the app — it shines from beside the lens, so the shadow hides under the scope.
- Tape the paper to the table so it cannot shift.



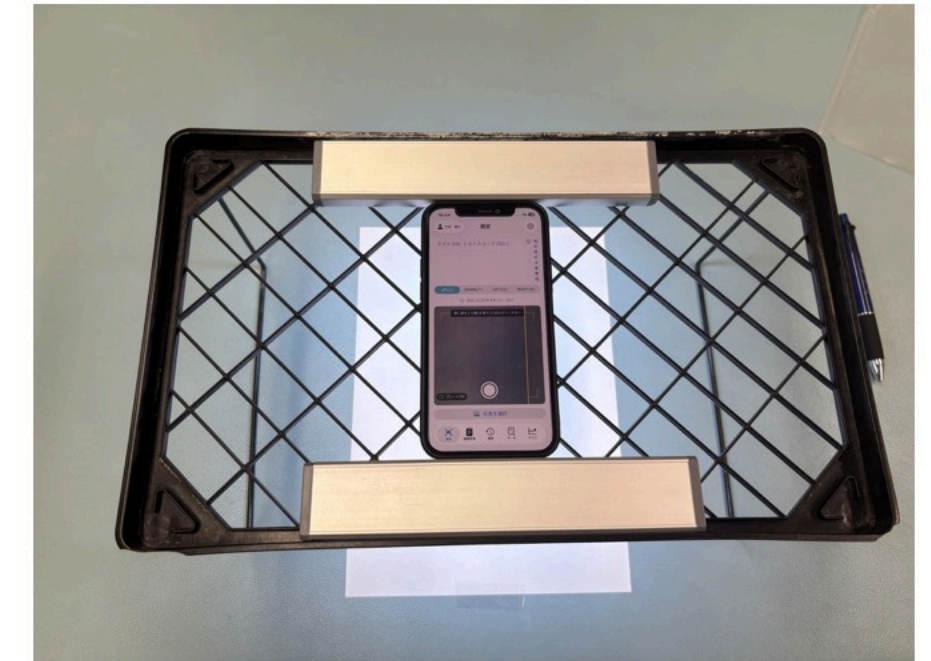
White A4 paper



Rack with a holed top



A simple light



All set up

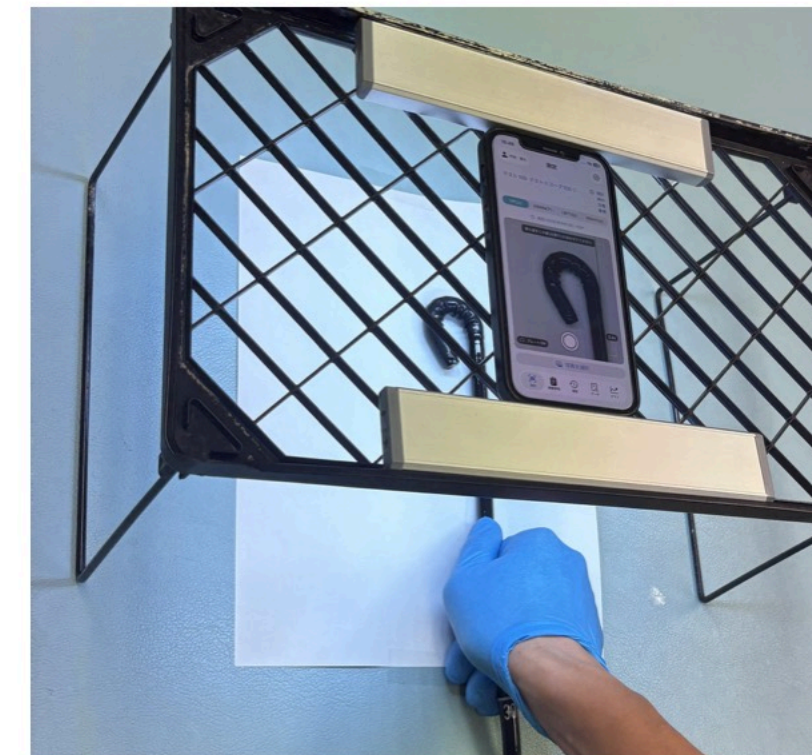
STAND

Shooting on a Stand (recommended)

- Let the camera look through a hole in the rack's top and shoot with the phone resting there — no shake, and the same height and angle every time.
- 1. Tape the A4 paper to the table → 2. Stand the rack over the paper
- 3. Rest the phone so the camera sits over a hole → 4. Lay the scope on the paper and shoot, tapping “Light On” if a shadow falls on it



Phone resting on the rack



Shooting the scope

Shooting Handheld

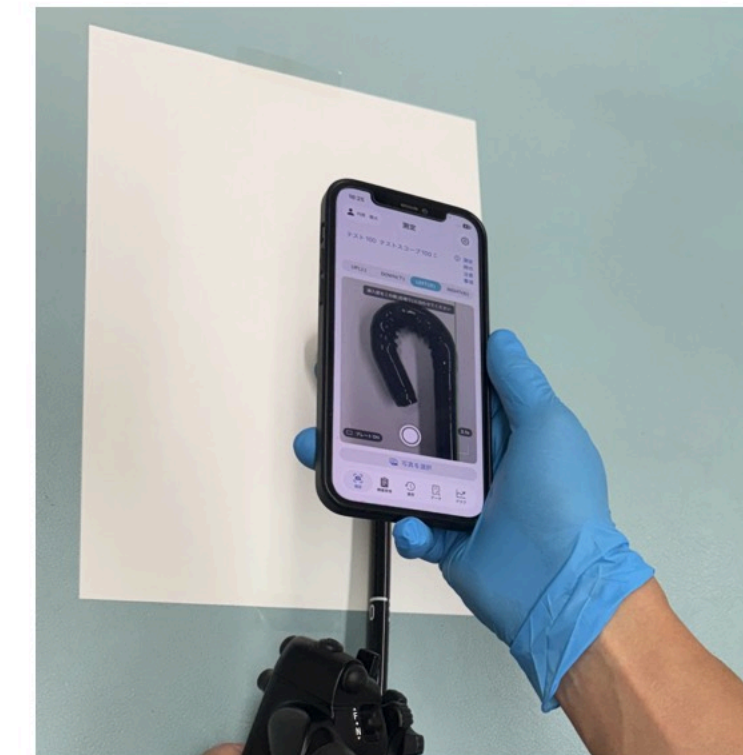
- If you cannot set up a stand, holding the phone by hand still works.
- Settings (gear) → turn on “Handheld mode” to show a level indicator. Centre the bubble and you are shooting from directly above.
- [Important] Rest the scope on the paper. Held up off it, the scope photographs smaller than it is and the angle comes out wrong.



Good — resting on the paper



Bad — the tip is lifted



By hand: hold it directly above

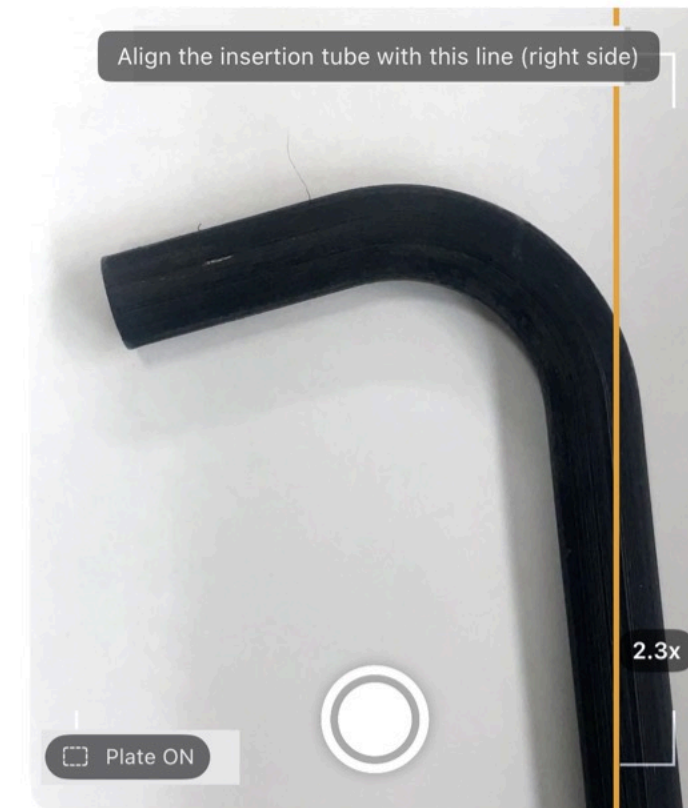
STEP 2

Taking the Photo

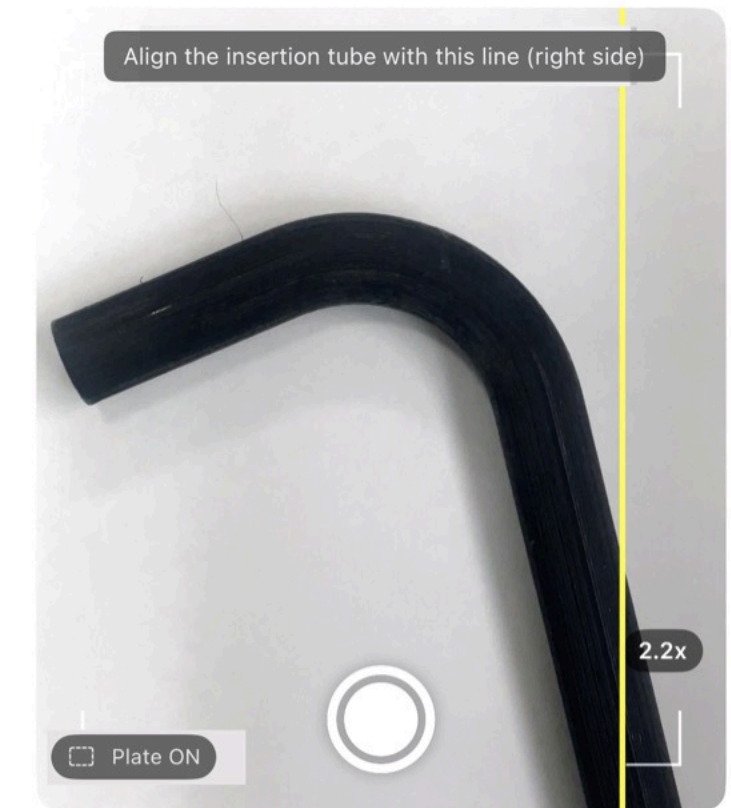
- Lay a plain white background (white A4 paper works well) underneath. A printed or woven surface — an angle plate, non-woven cloth, a wood-grain desk — causes misdetection.
- Rest the scope flat on the paper. Do not hold it up off the paper or let it tilt.
- Line the outer edge of the insertion tube up with the reference line on screen (green = good, orange = slightly tilted, yellow = needs adjusting).
- The reference line shifts left or right depending on the direction, to leave room on the side the tip bends toward.
- Zoom until the tip and the base sit inside the corner guides, then shoot somewhere bright. If a shadow will not go away, tap “Light On” at the bottom right.
- Changing the direction after you shoot discards that photo.



Green = good



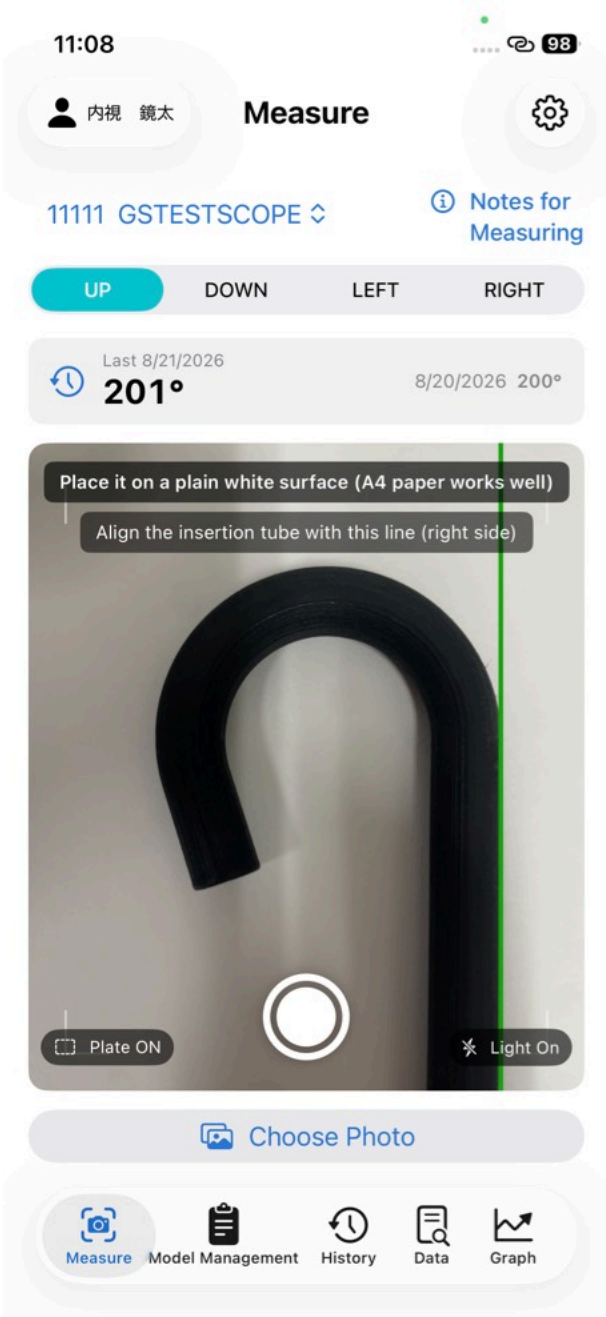
Orange = slightly tilted



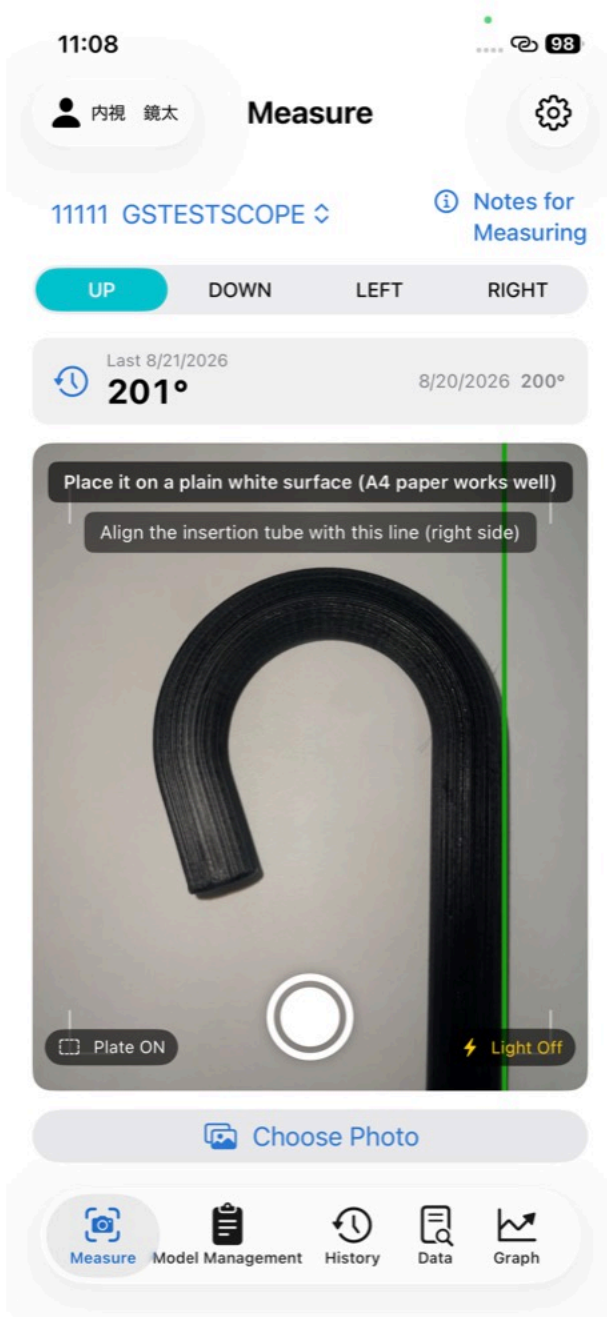
Yellow = needs adjusting

Buttons and Messages on the Shooting Screen

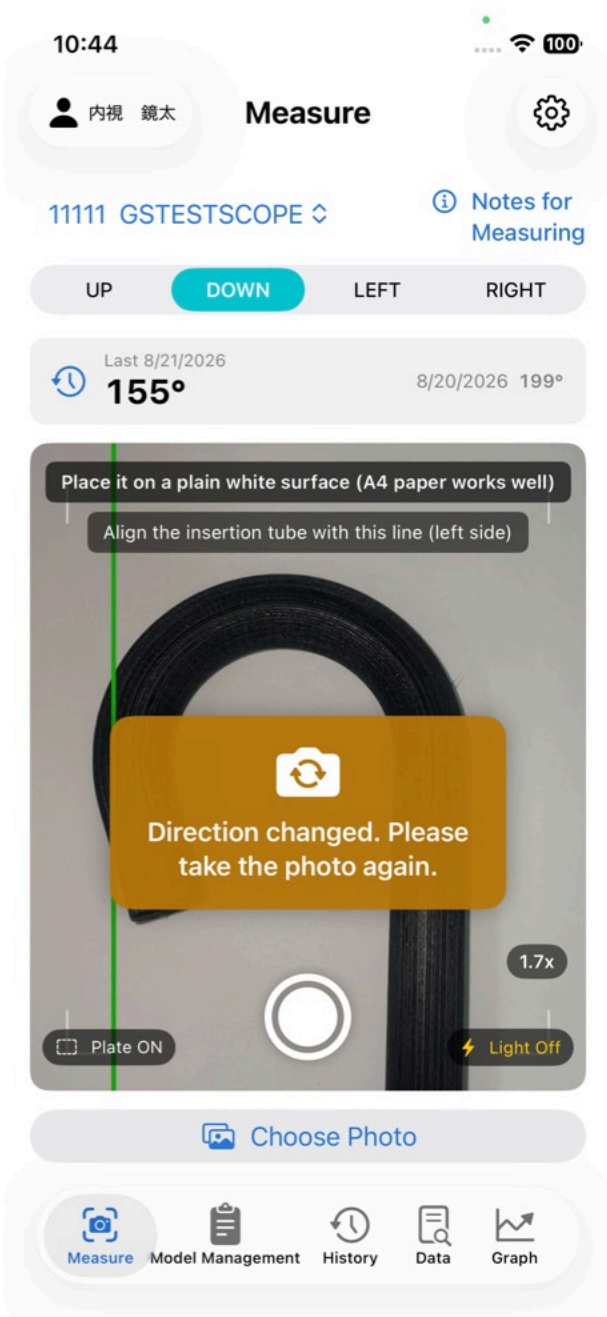
- Tap “Light On” at the bottom right to switch the phone’s light on. While it is on the button reads “Light Off” in yellow.
- The phone’s light sits right beside the lens, so the shadow hides underneath the scope. A dark shadow causes misdetection, so use this when a shadow will not go away.
- Saying “light” switches it too. It goes off by itself when you leave the shooting screen.
- Changing the direction after you shoot discards that photo and tells you so. Shoot again for each direction.



Off — tap to turn it on



On



After changing direction

STEP 3

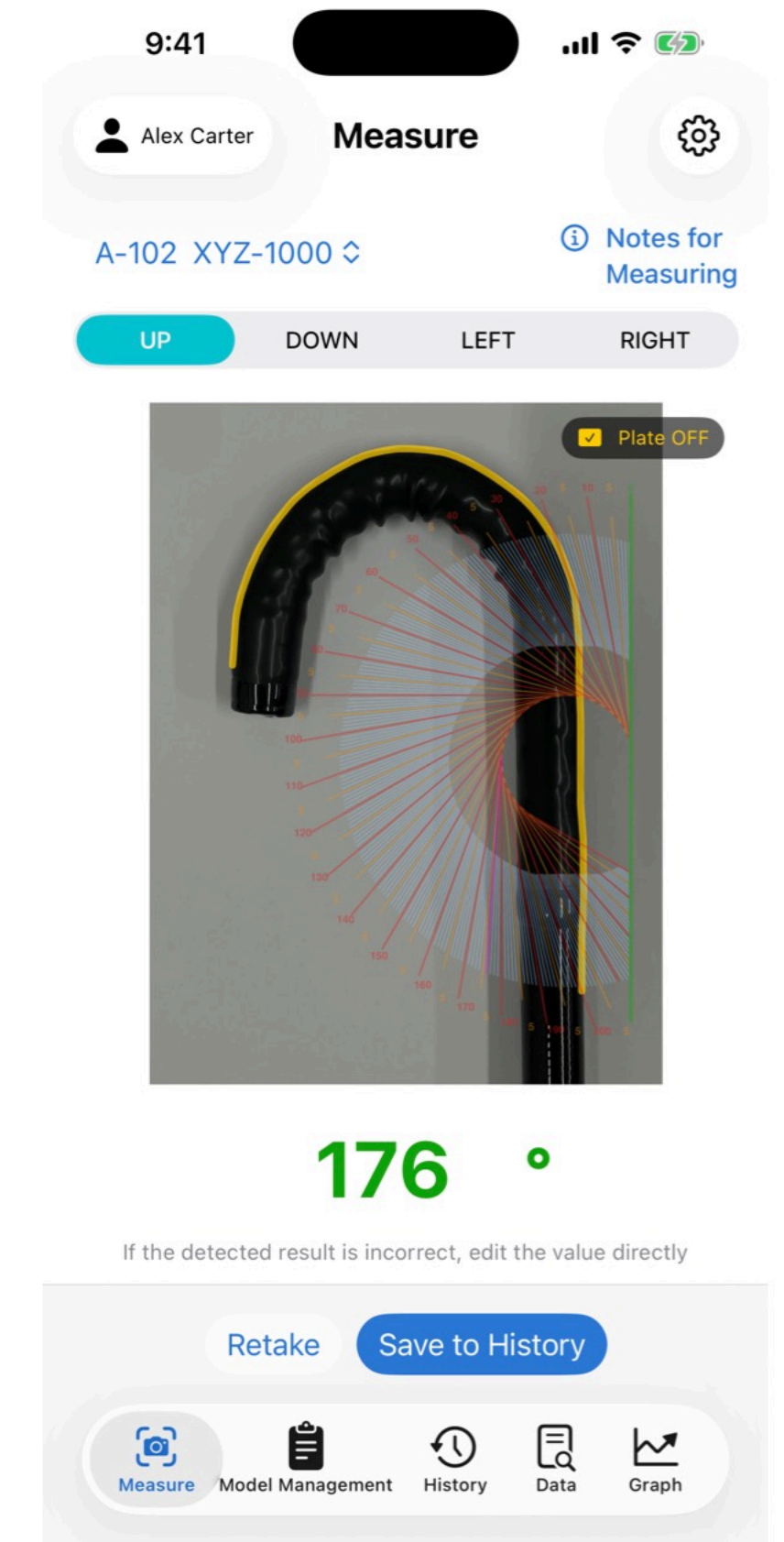
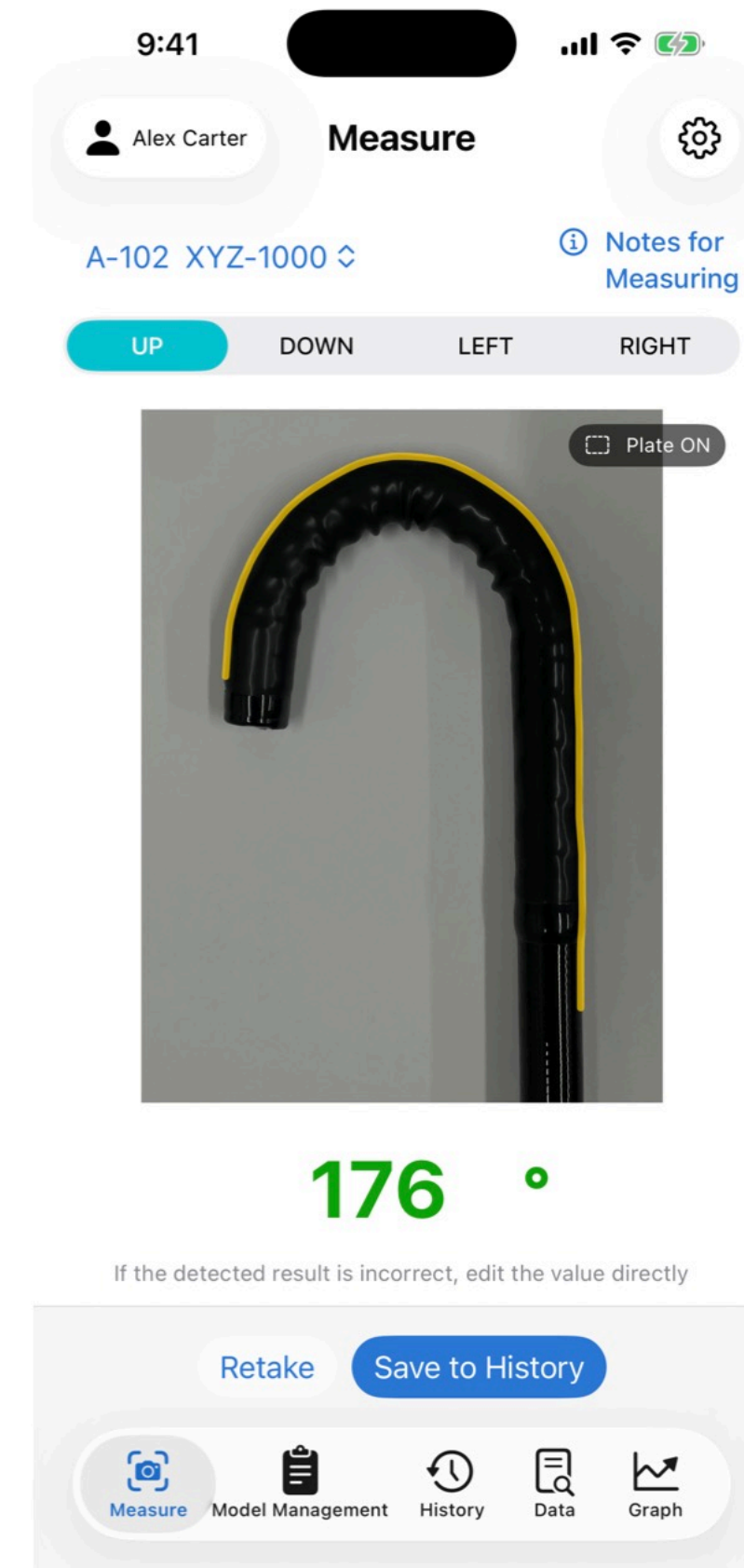
Checking and Saving After the Shot

- After a few seconds the angle is calculated automatically, and the measurement line (yellow) and the value appear.
- If the detection is wrong, tap the value and type the correct one.
- If the shaft is tilted a lot (more than 15°), a warning suggests retaking the photo.
- Check it over and tap “Save to History” (saying “save” out loud works too).
- To shoot again, tap “Retake”.



Overlaying the Angle Plate to Check

- 1. After shooting, tap “Plate ON” at the top right of the photo.
- 2. Drag with one finger to line the centre of the plate up with the base of the scope.
- 3. Pinch with two fingers to match the scale to the real thing (double-tap to reset).
- Only the half of the scale you need for that direction is shown. The line for the measured angle is highlighted in pink.
- * The on-screen plate is for reference only. It does not replace checking with the real angle plate.

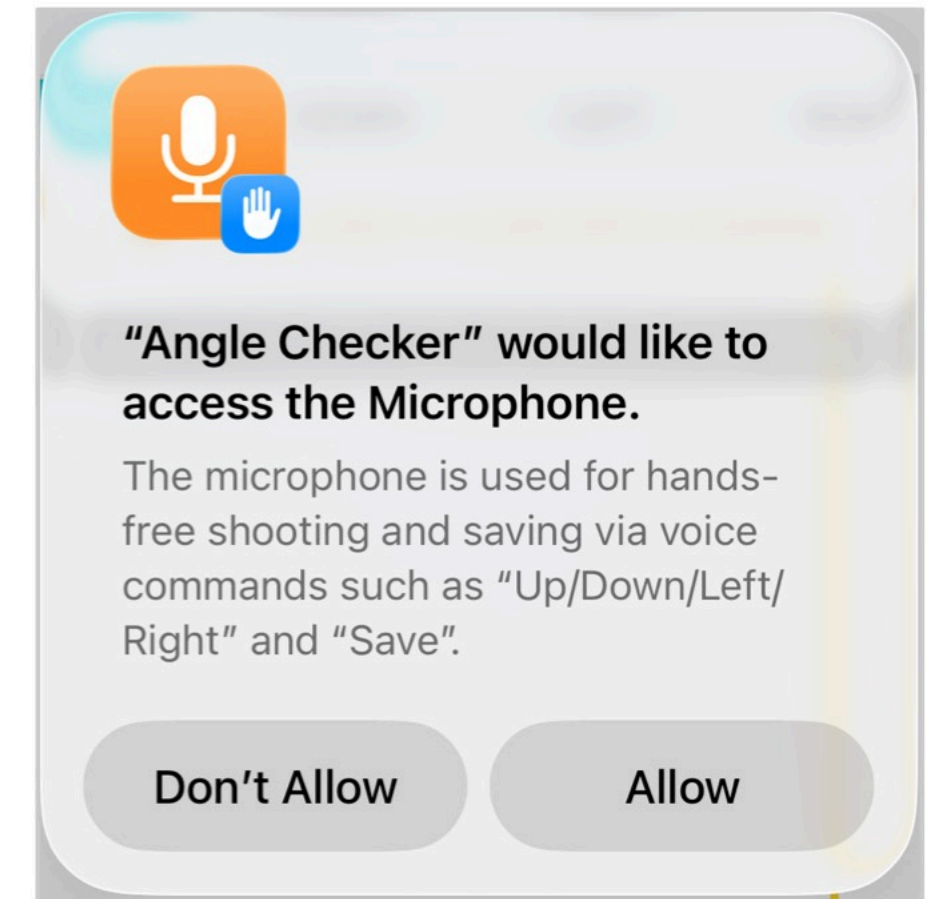
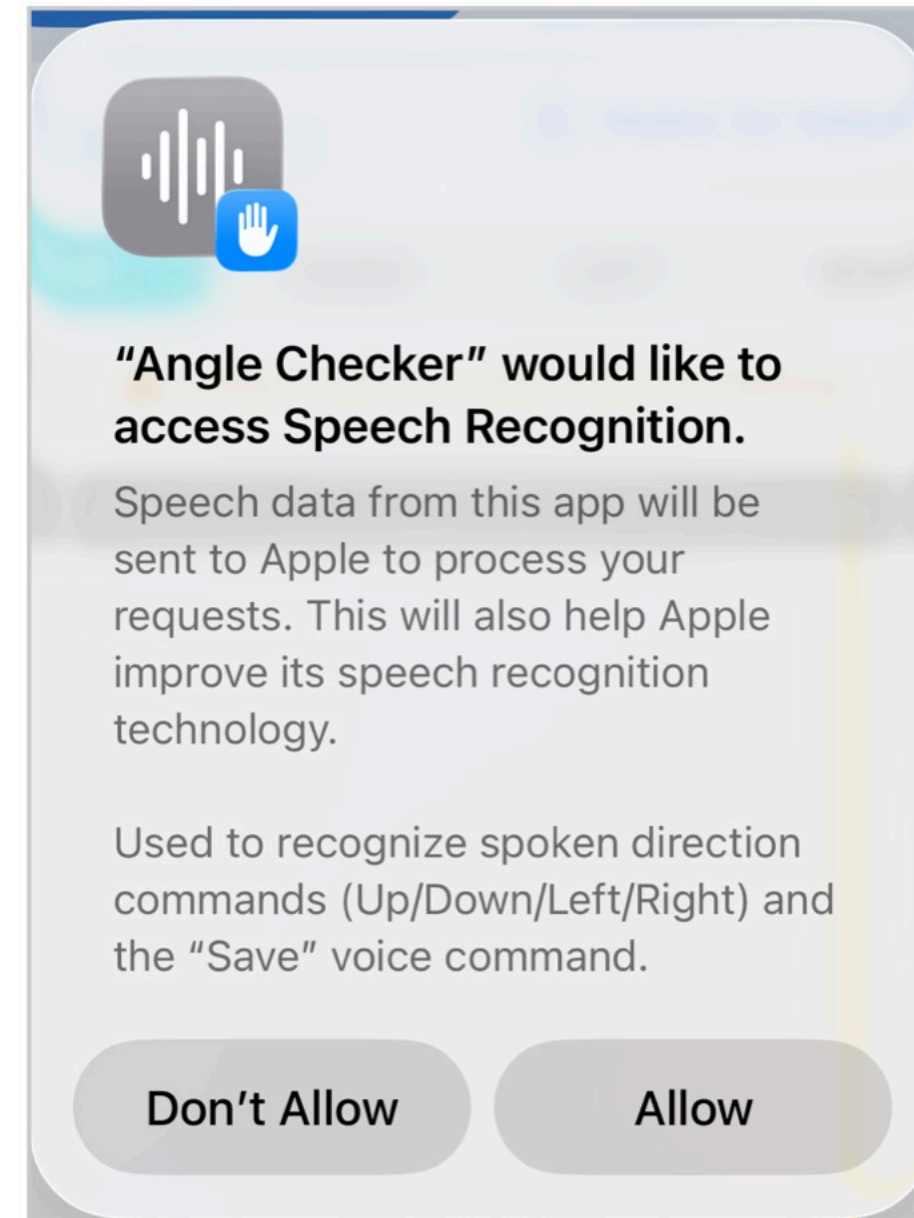


06

Using Voice Commands

Permissions on First Use

- To use voice commands, two permissions are needed the first time.
- 1. Access to speech recognition
- Used to recognise spoken directions (up / down / left / right) and commands such as “save”.
- 2. Access to the microphone
- Used to pick up your voice.
- Please choose “Allow” for both.



The Voice Commands You Can Use

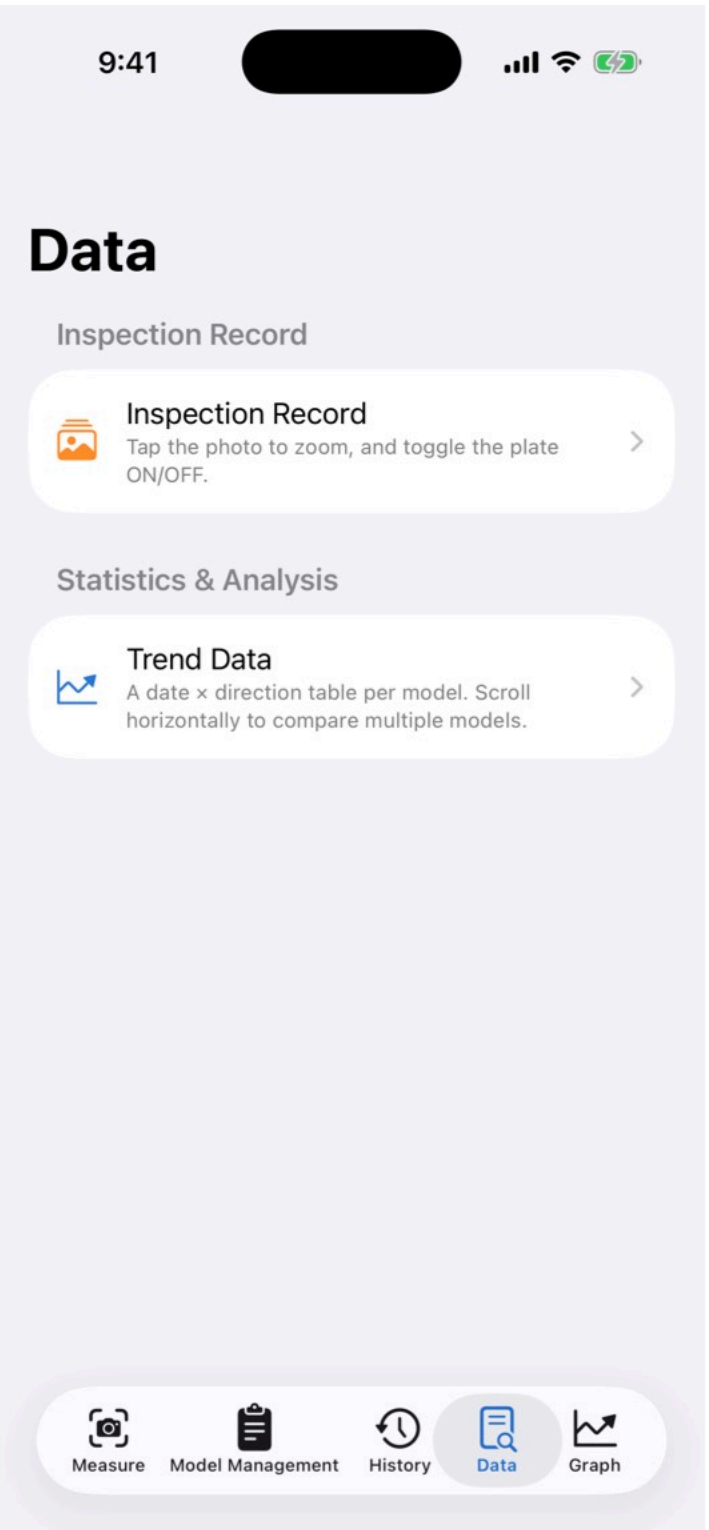
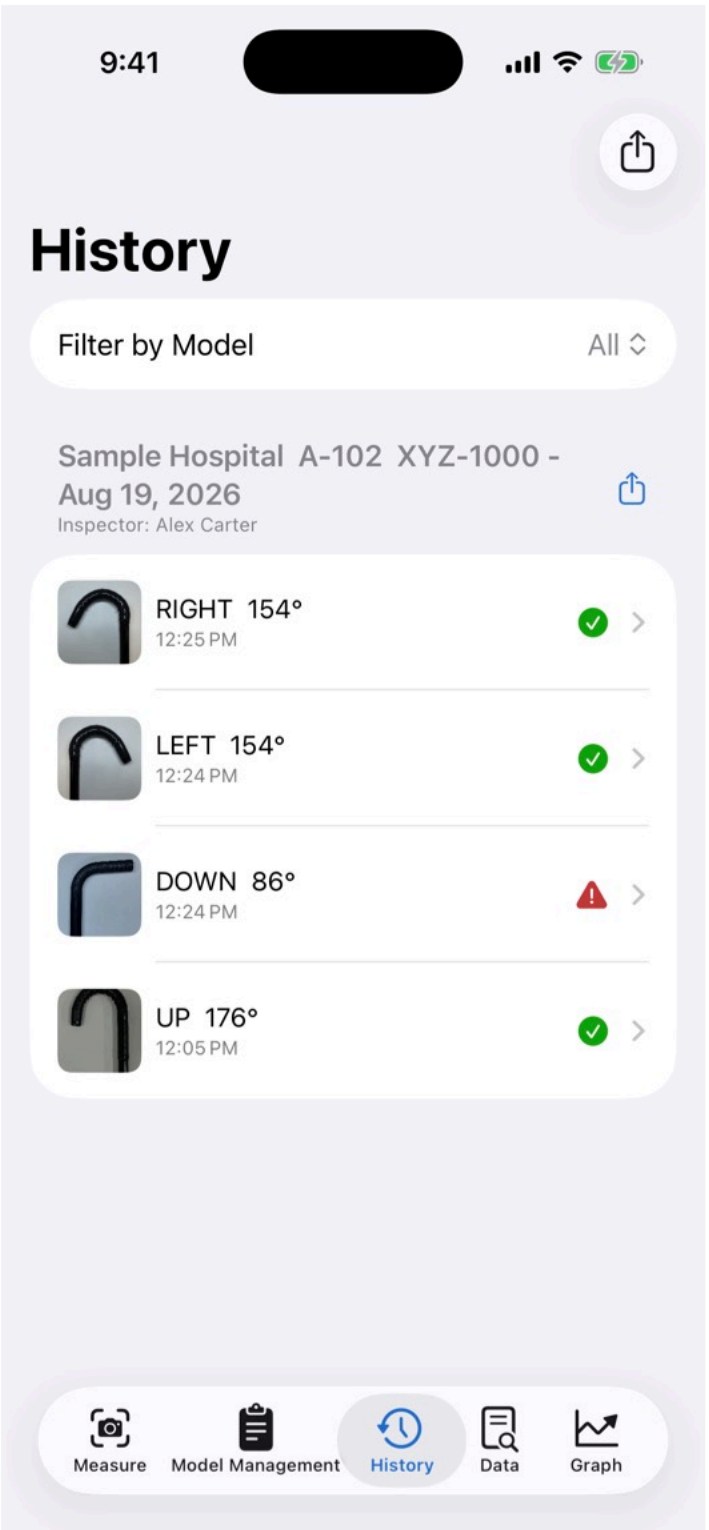
- Made for inspection work where both hands are busy: you can work by voice alone, without touching the screen.
- Voice control works only while the “Measure” tab is open.
- Change direction: “up” “down” “left” “right”
- Take a photo: “capture” “shoot” “photo”
- Save: “save”
- Shoot again: “back” “retry” “redo” “cancel”
- Zoom: “zoom in” “closer” / “zoom out” “farther”
- Focus: “focus”
- Light: “light” “lamp”

07

Records and Reports

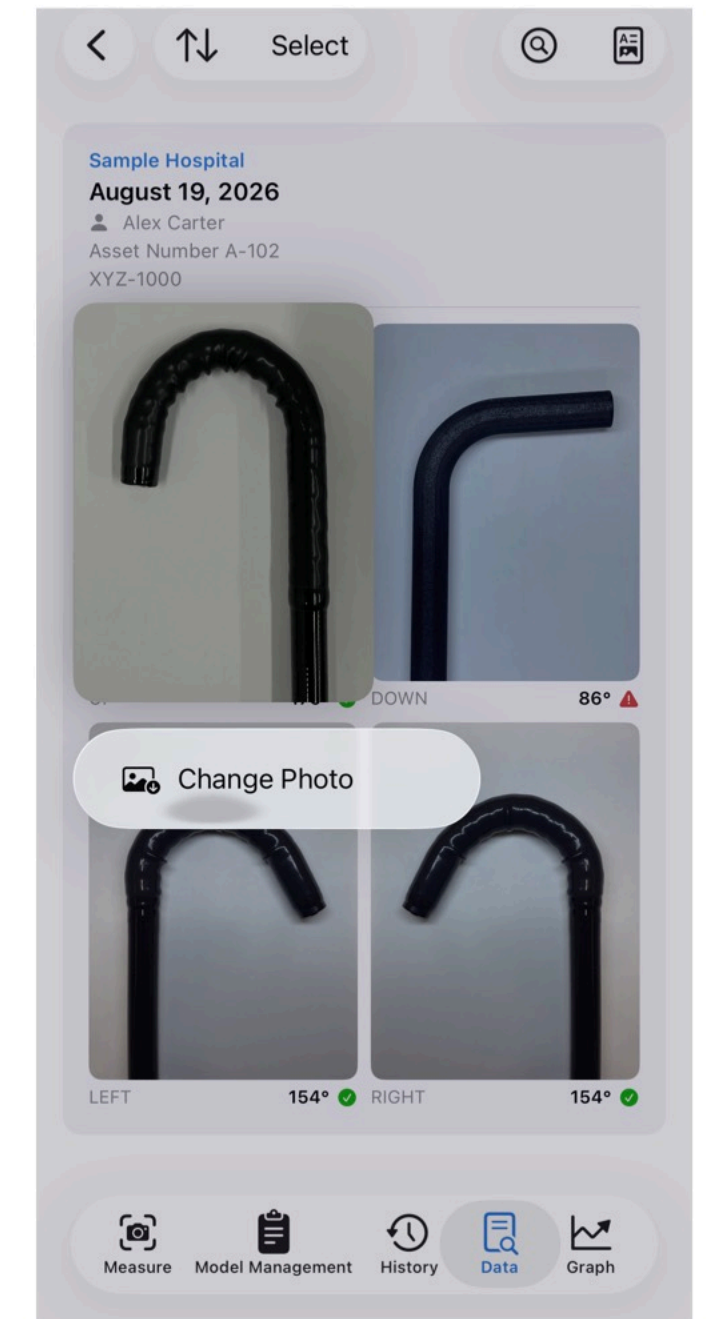
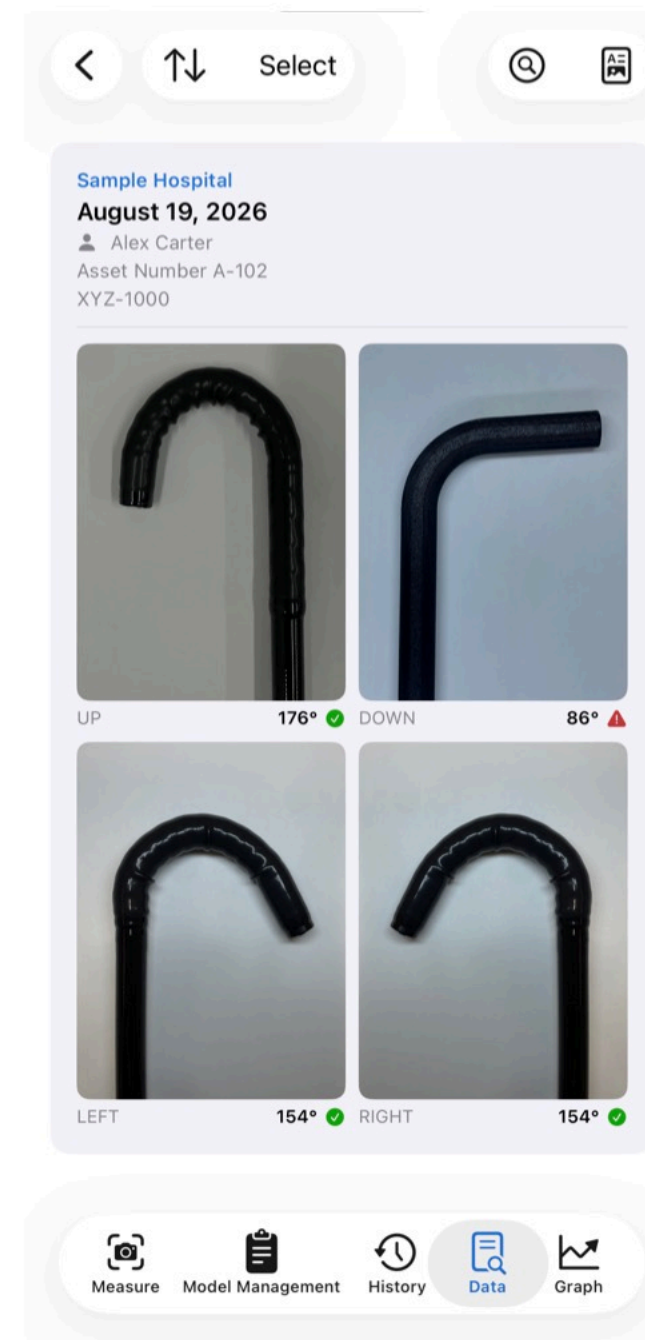
History / Data / Graph Tabs

- History tab: check, correct, and delete day-to-day records (for use on site)
- Data tab: browse and sort everything, and build reports (for monthly reporting and audits)
- Graph tab: see how the angle changes over time as a line graph. If the specified angle and tolerance are registered, the passing range is shown as a band.



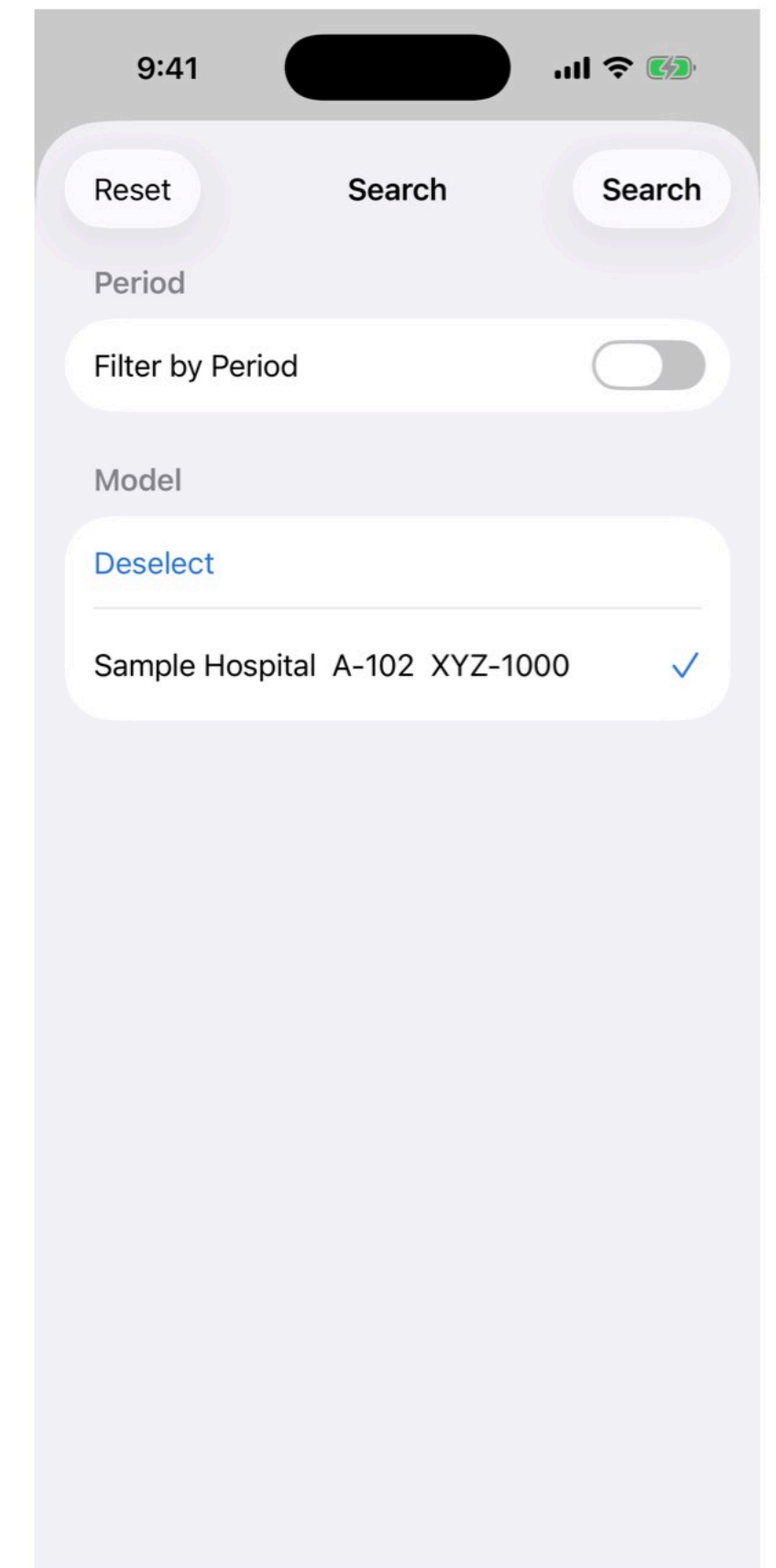
Viewing Records and Swapping Photos

- Data tab → “Inspection Record” lists the four directions' photos and angles for each model and date.
- Tap a photo to enlarge it (you can overlay the plate here too).
- Press and hold a photo → “Change Photo” to swap in another shot of the same direction.
- The candidates are listed with their time and measured angle, so you can pick the best one.



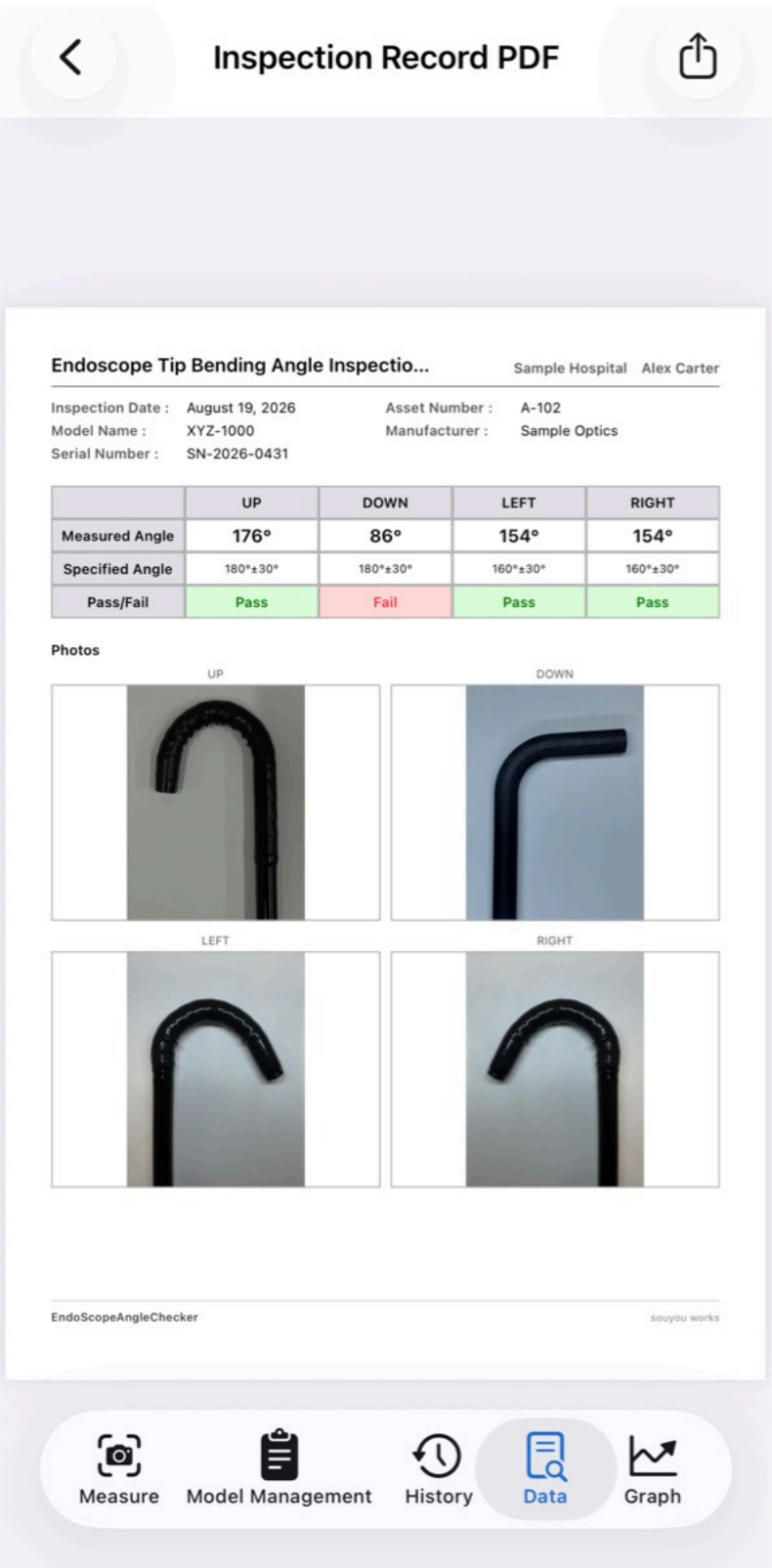
Searching and Filtering Records

- Use the search icon at the top of the inspection record list to filter.
- “Filter by Period”: set a range of inspection dates.
- “Model”: show only one model (by hospital and asset number).
- For monthly reports and audits, you can pull out just the records you need.



Exporting Reports as PDF and CSV

- Inspection Record PDF: one page per model and date, with a table of angle, specified angle, and pass/fail, plus the photos for all four directions
- Inspection Record CSV: one row per day, with the asset number, model name, and all four angles side by side
- Trend CSV: the four directions stacked vertically for each date (also ideal as a backup)
- Exporting opens the share sheet, so you can email it, AirDrop it, or save it to Files.
- As a regular backup, we recommend saving the Trend CSV or the Inspection Record PDF.

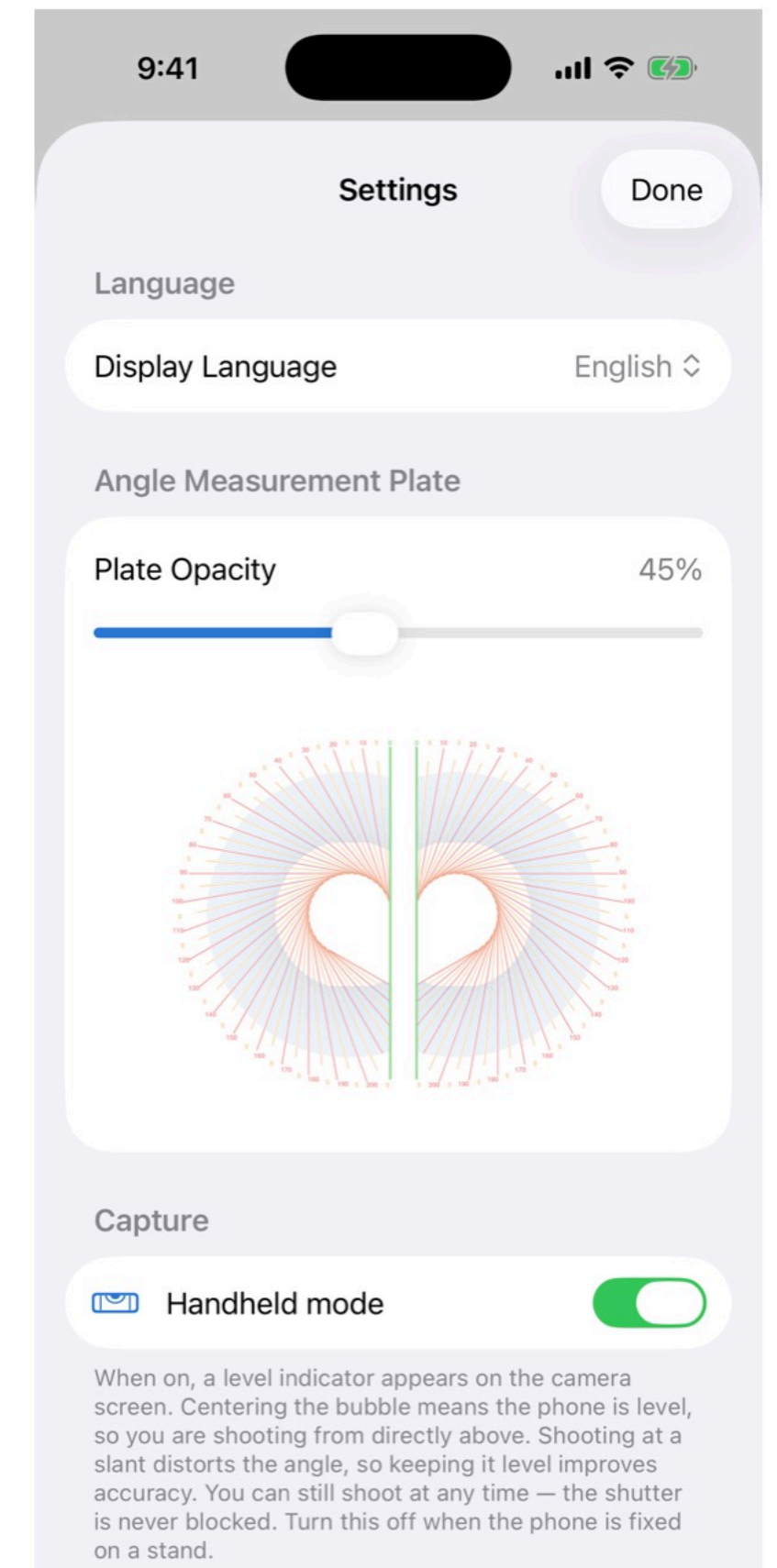


08

Settings

The Settings Screen

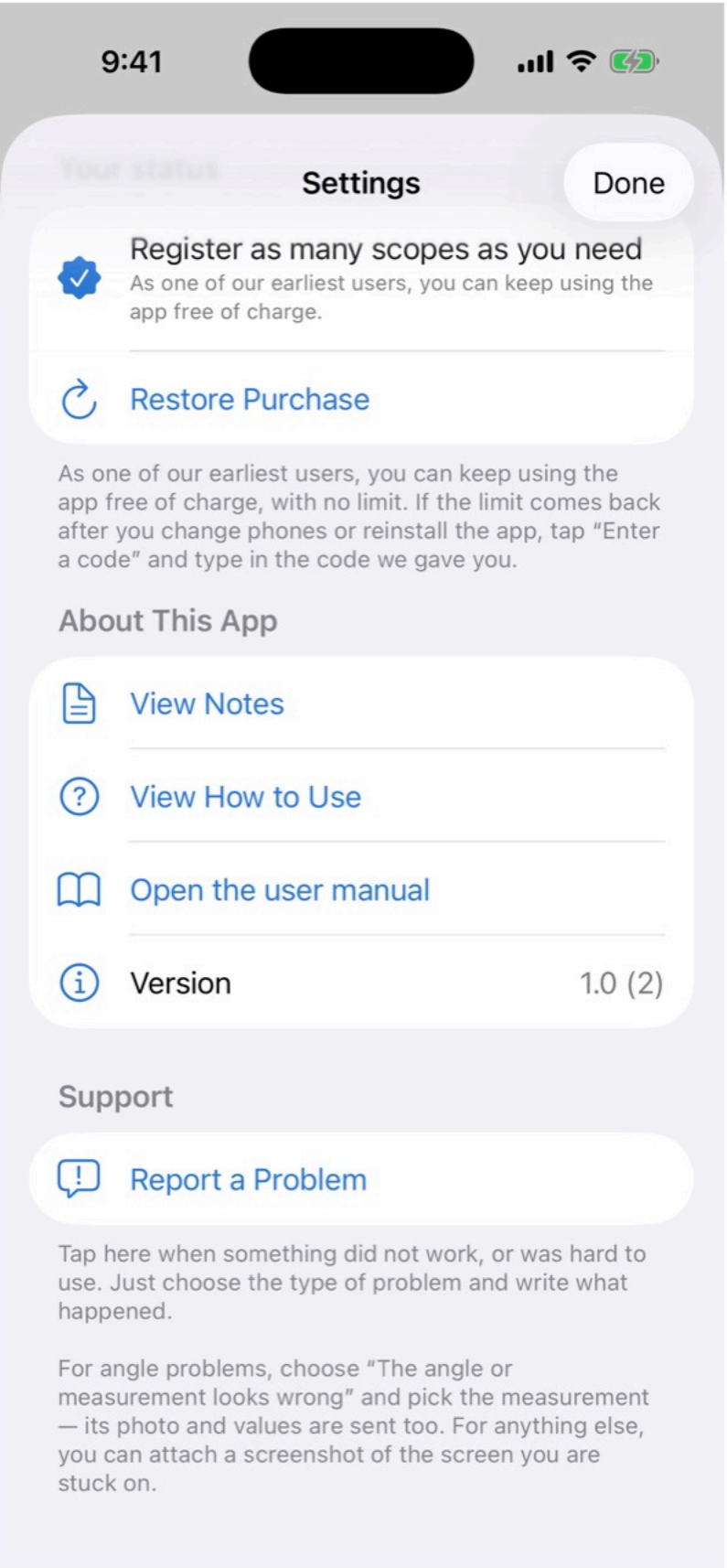
- Open it from the gear icon at the top right of the “Measure” tab.
- Display Language: 日本語 / English / 中文(简体)
- Plate Opacity: how strongly the angle plate is drawn (10%–90%)
- Handheld mode: turn it on to show a level indicator.
- Management: Models / Manage Hospitals / Manage Inspectors / Transfer Data



NOTE

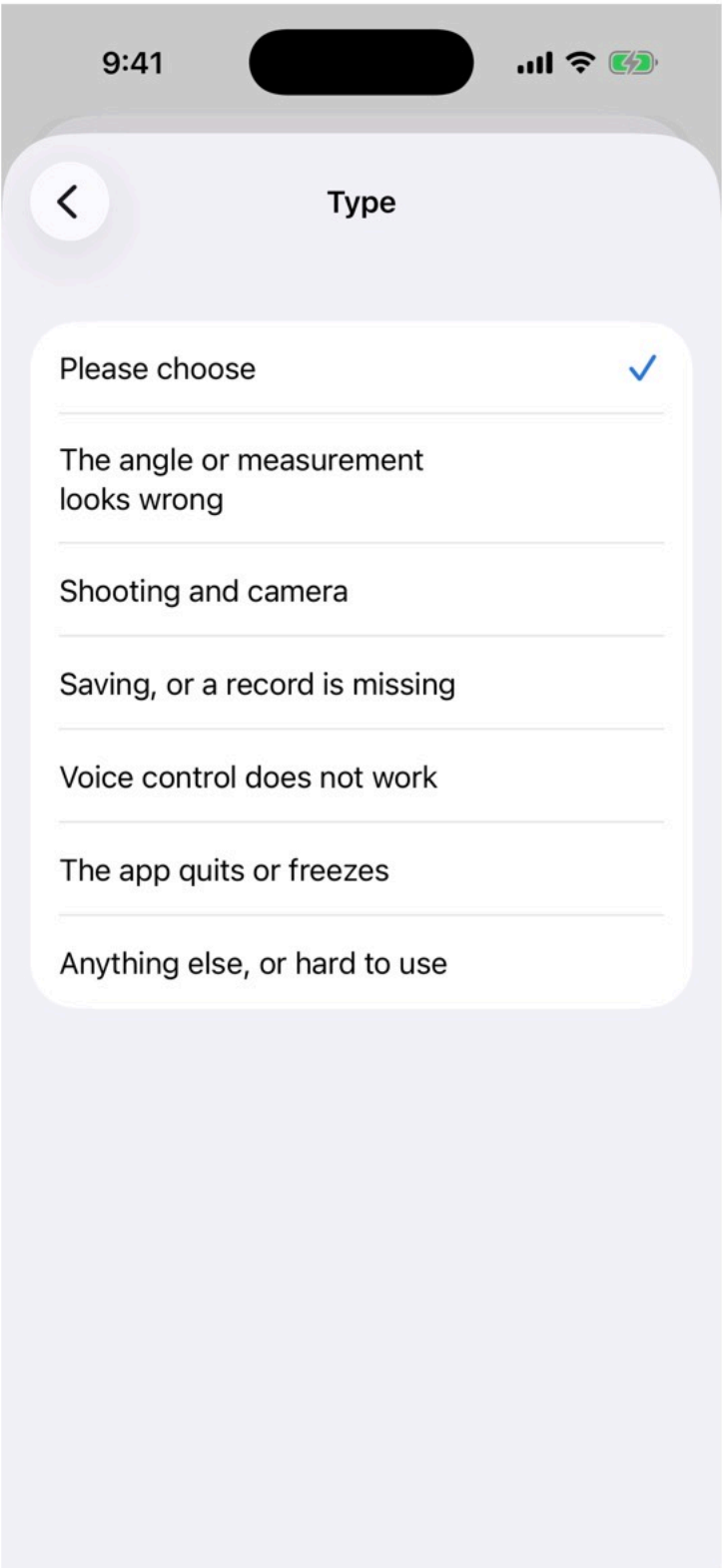
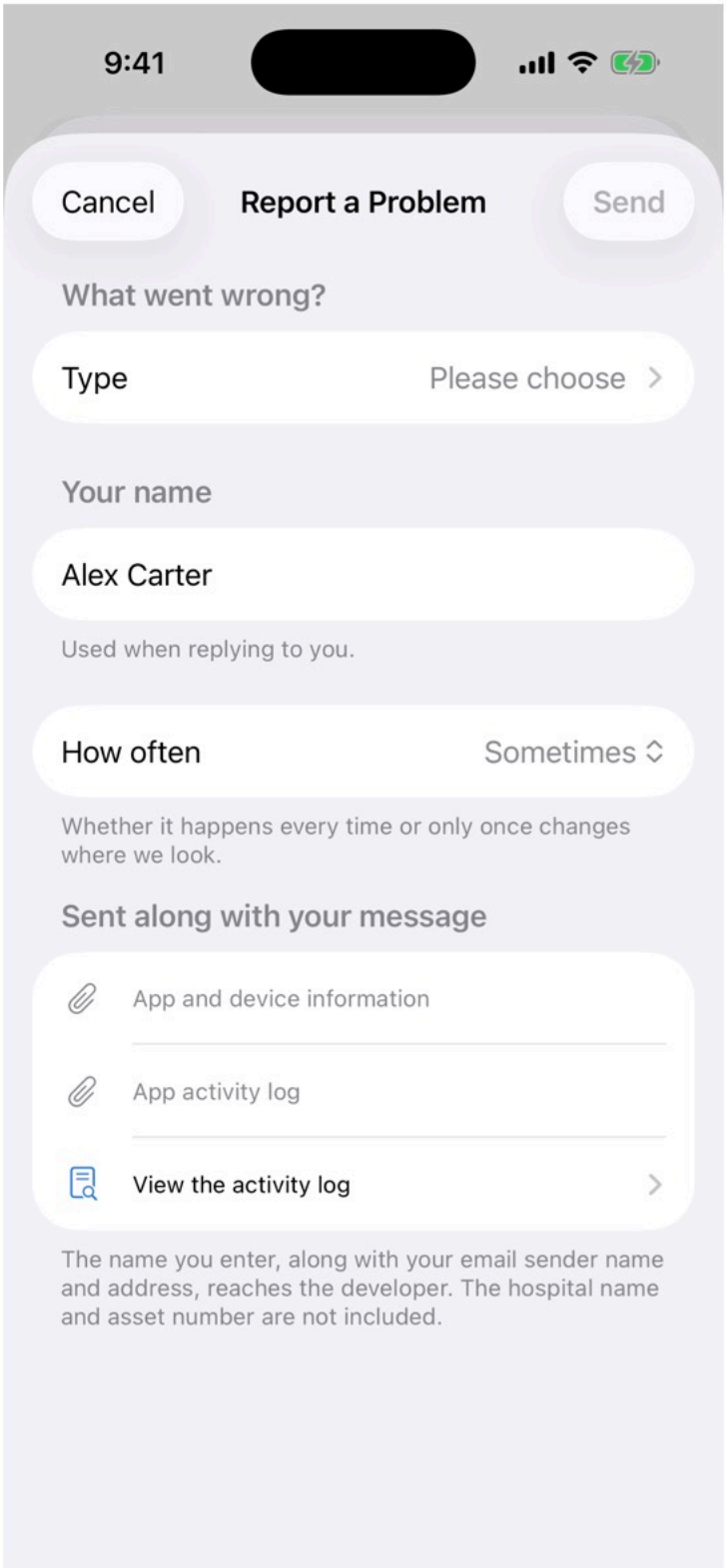
Other Settings

- About This App: open the notes, the how-to tutorial, and this manual at any time.
- Version: please quote this number when you report a problem.
- Report a Problem: choose a type, describe it, and add your name — the activity log is sent along with it.



Report a Problem

- Settings (gear) → “Report a Problem”. This is the only way in.
- 1. Choose a type (angle & measurement / camera / saving / voice / crashes / anything else).
- 2. Describe it. Examples of what to write appear below, for each type.
- 3. Enter your name. It is used when we reply.
- 4. Tap “Send”: a mail compose screen opens — just send it.



How to Attach Photos, and What Gets Sent

- Choosing “The angle or measurement looks wrong” reveals “Choose a measurement”. Pick the one in question and its photo and values (measured angle, specified angle, direction, model name) are sent too.
- If you would rather not send the photo, turn “Attach the measurement photo” off and report the values alone.
- For every other type you can attach a screenshot. To take one: press the side button and the volume up button together (on models with a Home button, press Home and the power button).
- What is sent: the type, what you wrote, your name, app and device information, and the app's activity log.
- What is not sent: hospital names, asset numbers, and your other measurement records.
- Because it goes by email, your email address and name do reach us (they are needed so we can reply).